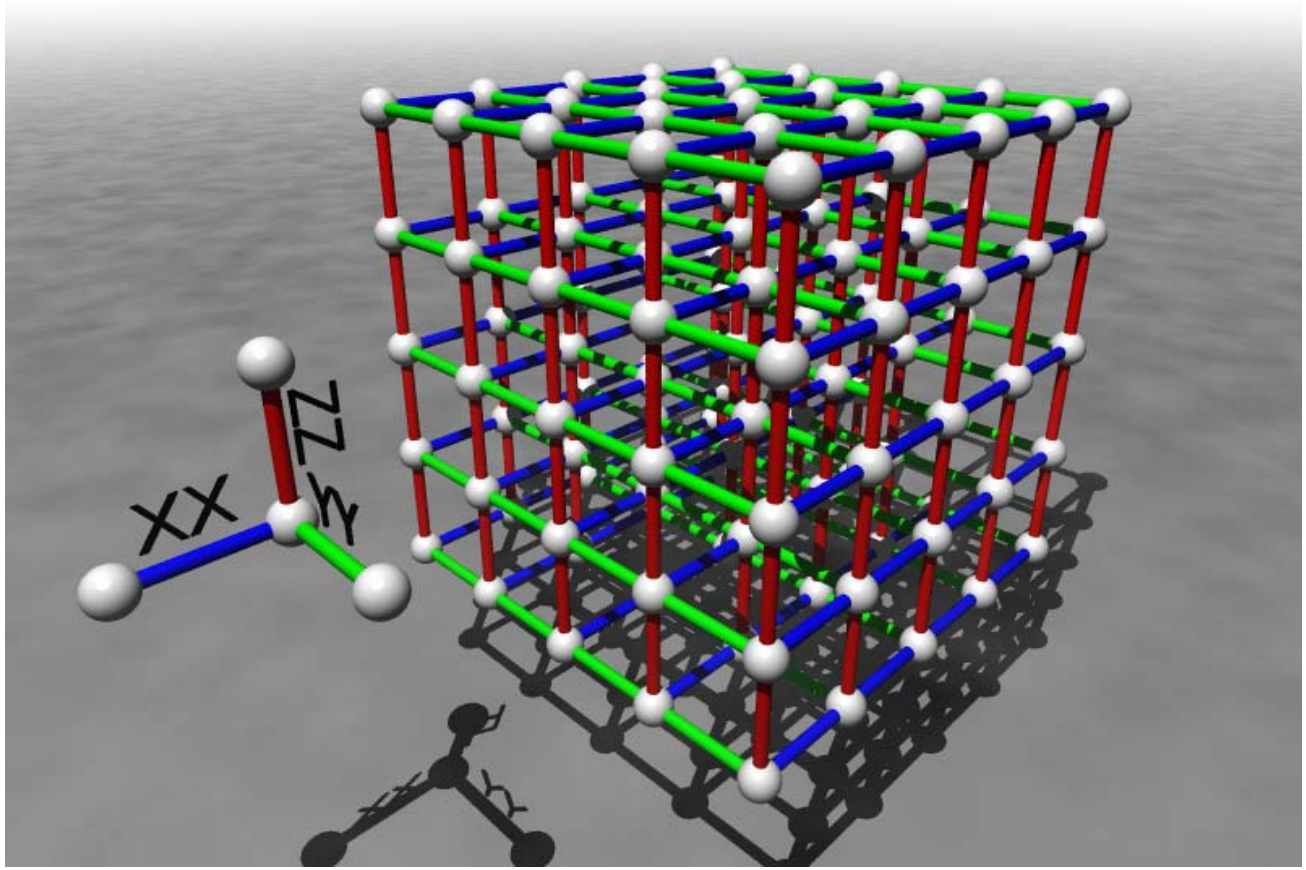
THS SNTNC HS N VWLS

Spatial Locality and Robust Quantum Computation



Dave Bacon
Department of Computer Science & Engineering
University of Washington

I am a Quantum Information Science Theorist, but…..

Trapped Ions Saved My Life

Experimental Demonstration

REPORTS

A Decoherence-Free Quantum Memory Using Trapped Ions

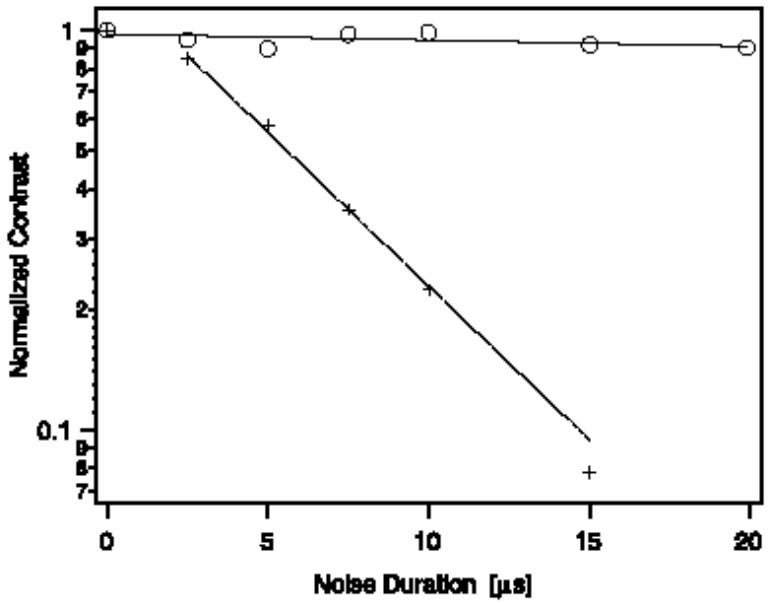
D. Kielpinski,[1]* V. Meyer,[1] M. A. Rowe,[1] C. A. Sackett,[1]
W. M. Itano,[1] C. Monroe,[2] D. J. Wineland[1]

SCIENCE VOL 291 9 FEBRUARY 2001

collective phase errors:

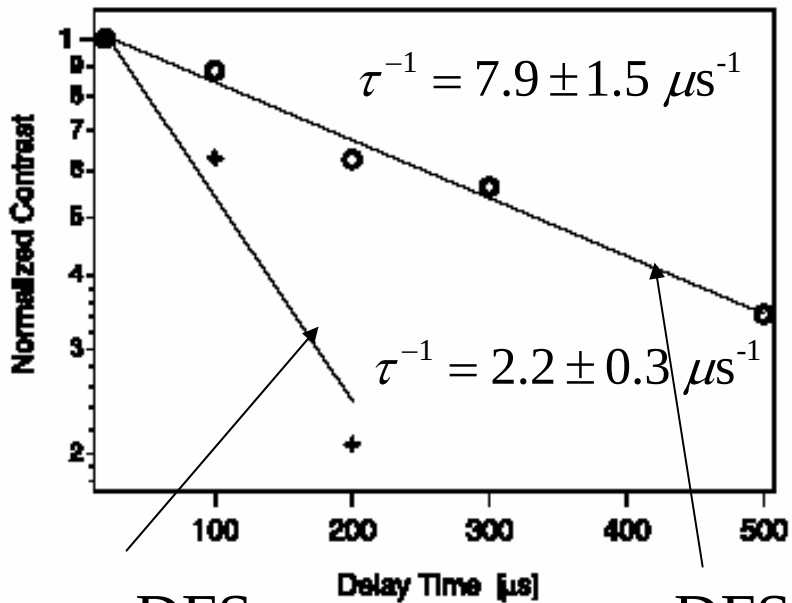
$$S_z^n = \sum_{i=1}^{n} s_z^{(i)}$$

engineered noise (detuned laser)



(see also Kwiat et al., Science 2000)

ambient conditions!



non-DFS DFS

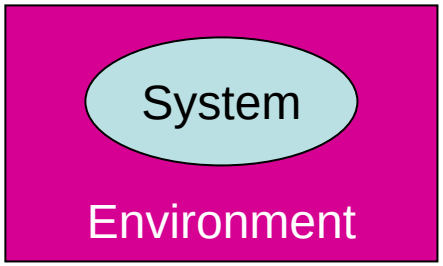
Building a Quantum Computer Is Full of Problems



Is a Quantum Computer Even A Valid Model of Computation?

The First Problem

THE DECOHERENCE PROBLEM



$$H = H_S \otimes I + I \otimes H_B + H_I$$

open quantum systems can become entangled with their environment

EXAMPLE:

evolve

$$(\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |0\rangle_E \rightarrow \alpha|0\rangle_S \otimes |0\rangle_E + \beta|1\rangle_S \otimes |1\rangle_E$$

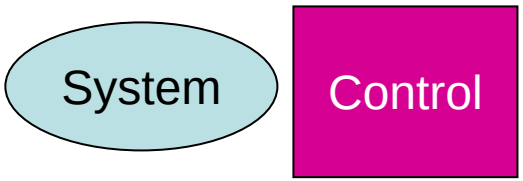
coherent superposition — evolve — incoherent sum

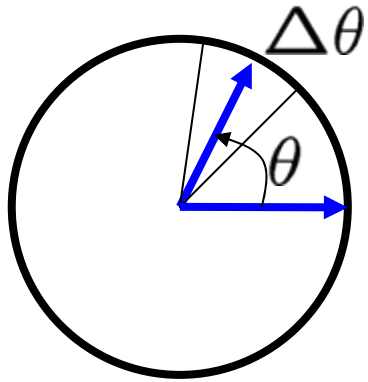
$$(\alpha|0\rangle_S + \beta|1\rangle_S) \rightarrow \rho = |\alpha|^2 |0\rangle_S\langle 0|_S + |\beta|^2 |1\rangle_S\langle 1|_S$$

1. Decoherence turns quantum information into classical information
2. It does this on a time scale which only a few hundred to million times slower than the rate we can manipulate the quantum information

Second Problem

THE FAULTY CONTROL PROBLEM

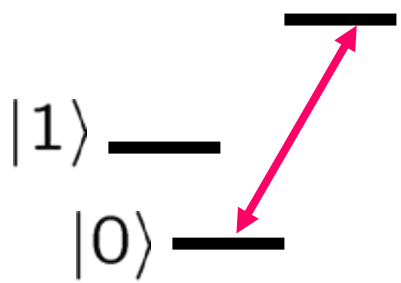




unitary gates cannot be implemented with arbitrary accuracy

THE FAULTY MEASUREMENT PROBLEM

measurements are imperfect

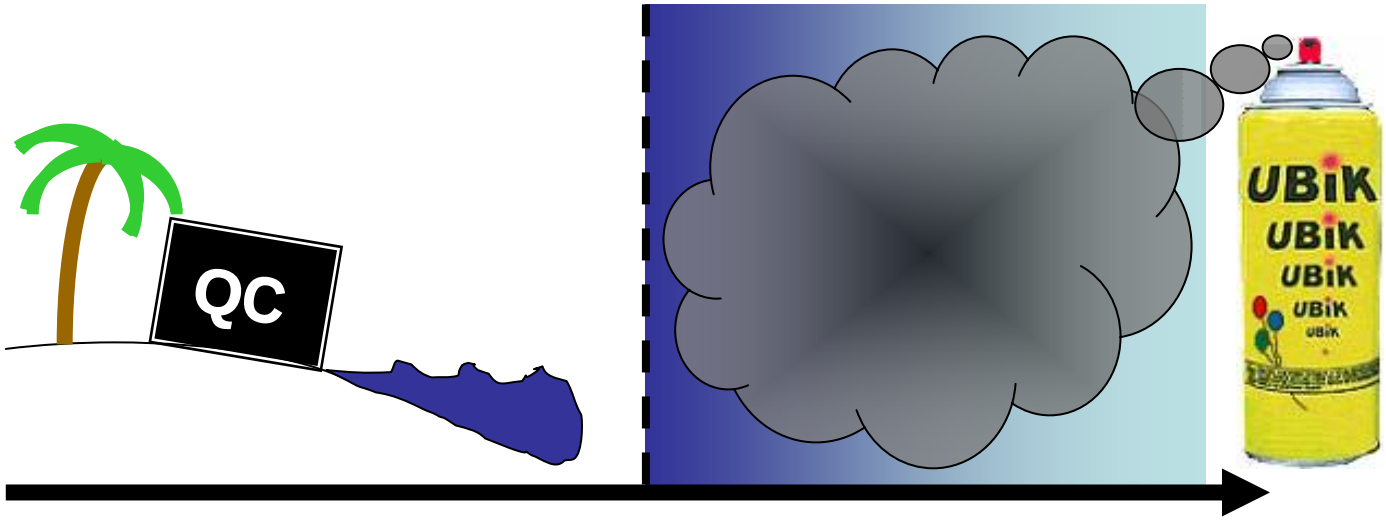


wrong result, wrong resulting state

related state preparation problem

LEAKAGE PROBLEM

Why We Are Here



“decoherence”

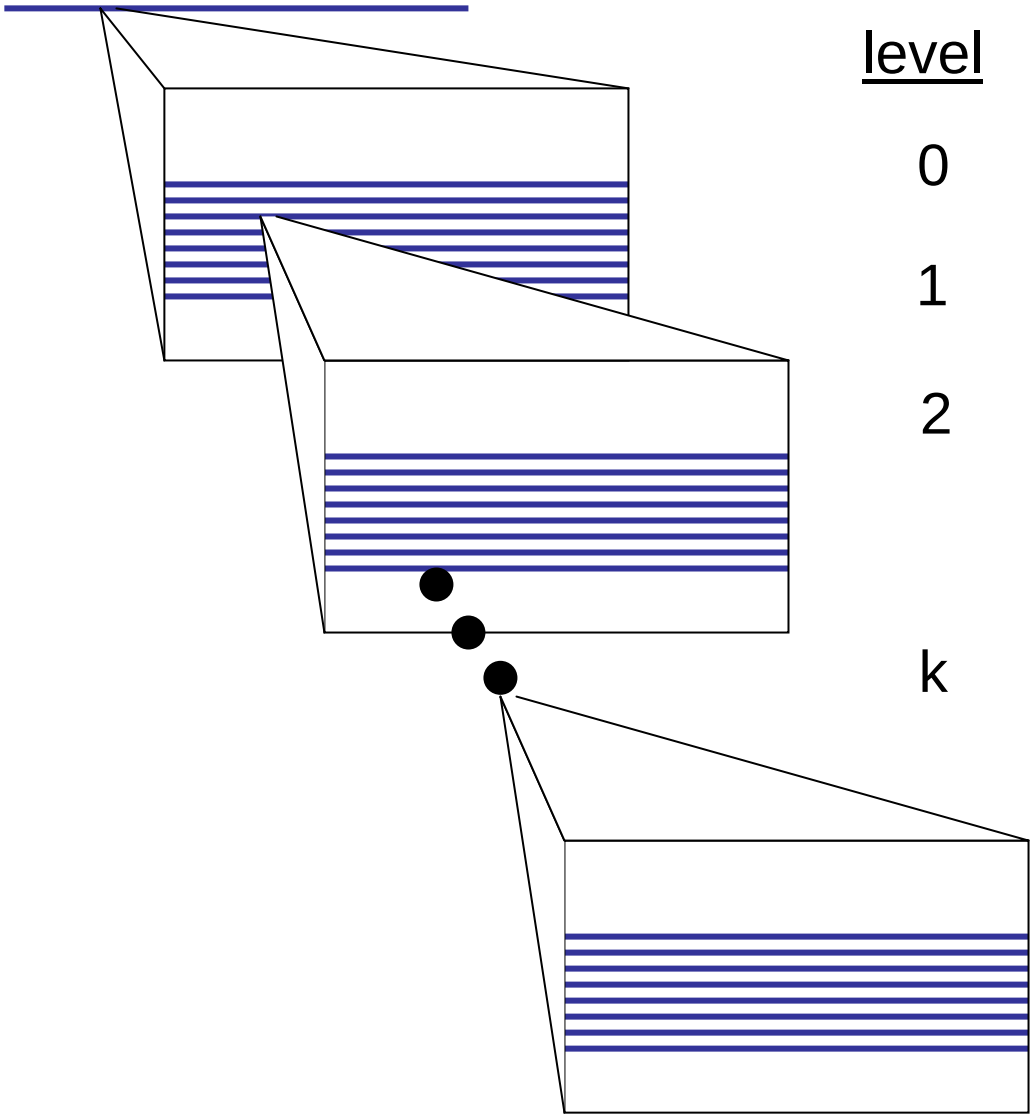
“imprecision”

Threshold Theorem for Fault-Tolerant Quantum Computation

[a partial list: Aharonov, Ben-Or, Kitaev, Knill, LaFlamme, Zurek, Shor, DiVincenzo, Gottesman, Preskill]

If decoherence is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

Concatenated Threshold Sketch



level	qubits	probability of failure
0	1 qubit	p
1	n qubits	cp^2
2	n^2 qubits	$c(cp^2)^2$
k	n^k qubits	$(cp)^{2^k}/c$

exponential decrease in # qubits if

$p<1/c=p_{thresh}$

Paths to the Threshold

1. CONCATENATED FAULT-TOLERANCE
2. NON-CONCATENATED FAULT-TOLERANCE ✔
3. NATURAL FAULT-TOLERANCE ✔



There is No Threshold Theorem

Threshold Theorems

TWO TYPES

provable and heuristic

ASSUMPTIONS

provably essential:

1. parallel operations
2. cold ancillas
3. error rates below 50% gate speed
4. ability to not leak or fix leaks

error model:

correlations (spatial and temporal)

decoherence types (single qubit, leakage, Markovian, etc.)

classical computation:

free fast robust classical computation?

architecture:

non-local gates, local gates

dimension of architecture

explicit architecture (scheduling, routing, etc.)

Local Threshold Theorem

Ground breaking:

Gottesman 1999 and Aharonov, Ben-Or 1999

Local threshold still exists

Local gates, but global classical computation

Dennis, Kitaev, Landahl, and Preskill 2001

surface quantum codes 2D (memory)
and 3D (computation)

memory threshold: 0.01

fault tolerant threshold: ~0.0001

Local gates, concatenation

Svore, Terhal, DiVincenzo 2004

2D Steane code, loss of a factor of 10-50.

Szkopek et al. 2004

1D Steane code, loss of a factor of 200.

Spatial Locality

This talk:

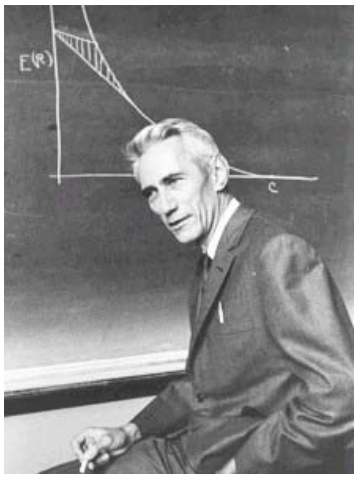
A Story of My Attempt to come to Grips with Locality

And

Adventures in where these Ideas Led

To Build a ~~Quantum~~ Computer Classical

Problem (1938): How to build a reliable system from unreliable components?



Claude Shannon

"A mathematical theory of communication" (1948)

How to communicate in the presence of noise

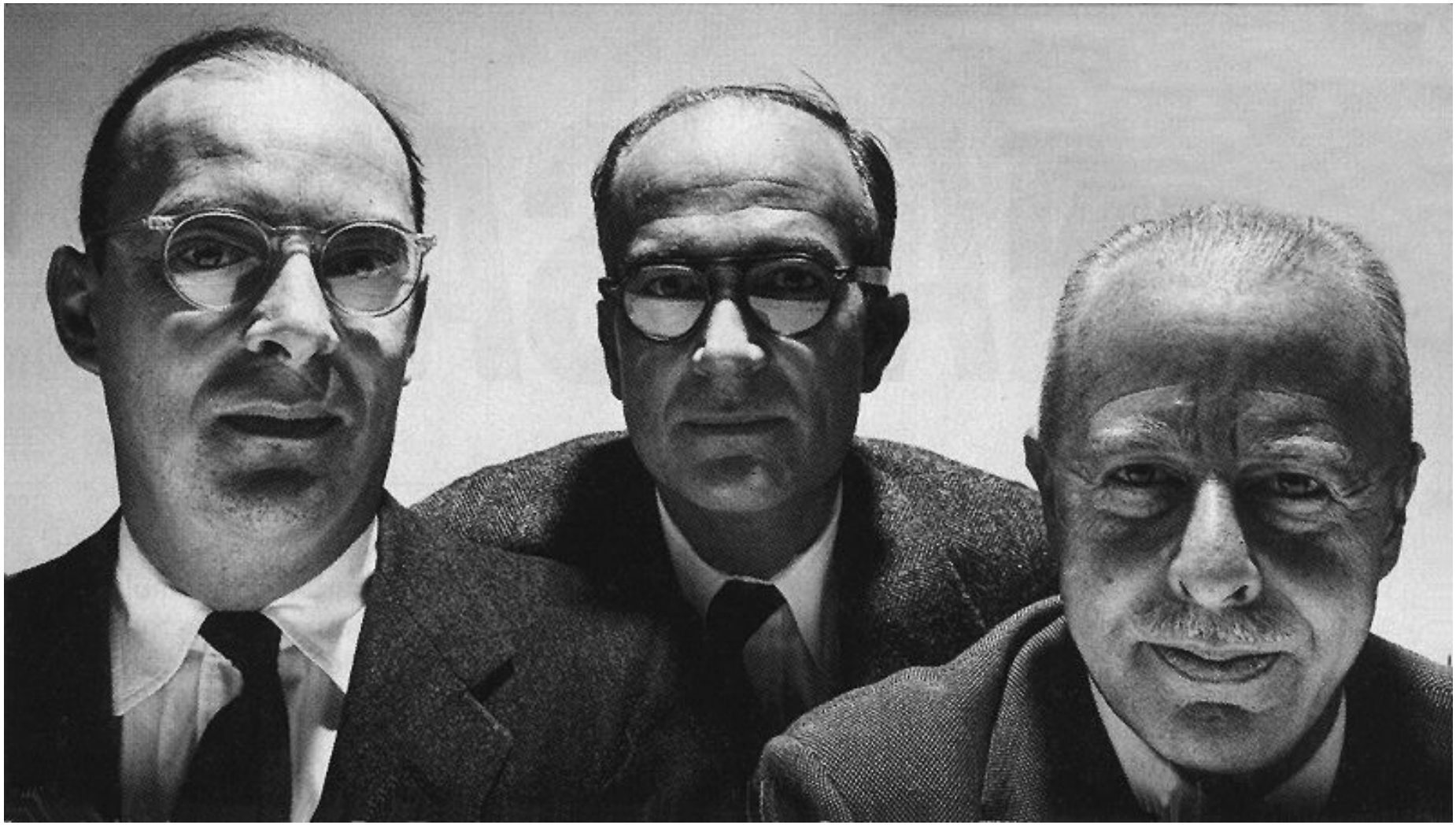
"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)



John von Neumann

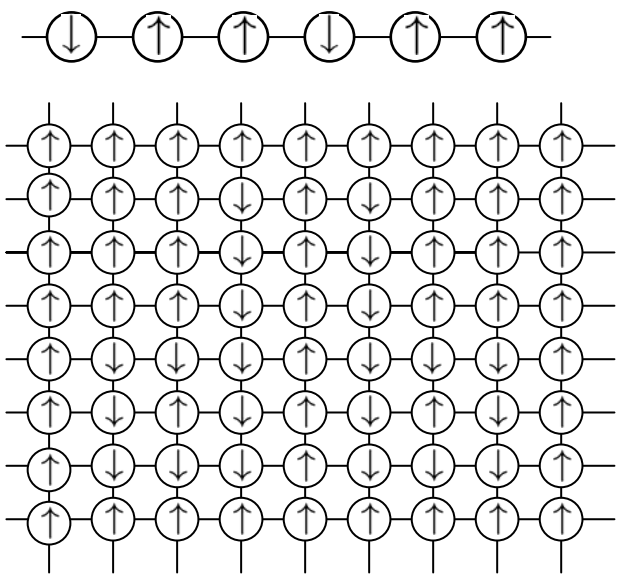
How to compute when the subsystems of your computer can fail?

A Different Path



Two Storage Testbeds

Ferromagnetic Ising models:

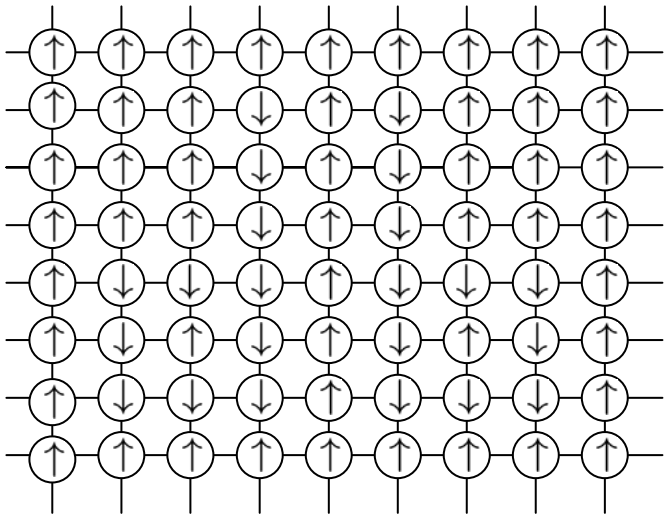


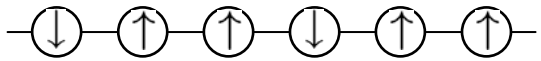
A. one dimensional

B. two (and >) dimensional

Ferromagnetic Ising Models







Ising model

spins on edges of a lattice: $s_i \in \{+1, -1\}$

energy of a configuration:

$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i \qquad J > 0$$

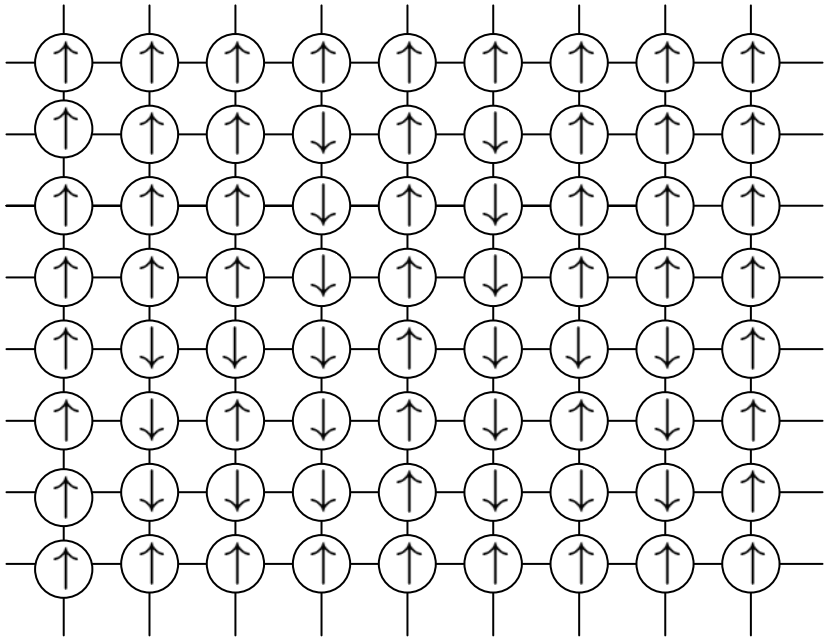
sum over neighbors

external magnetic field

$h = 0$ for now

Dynamic Ising Models





Dynamic Ising model (Metropolis update)

Interaction with environment causes spin flips

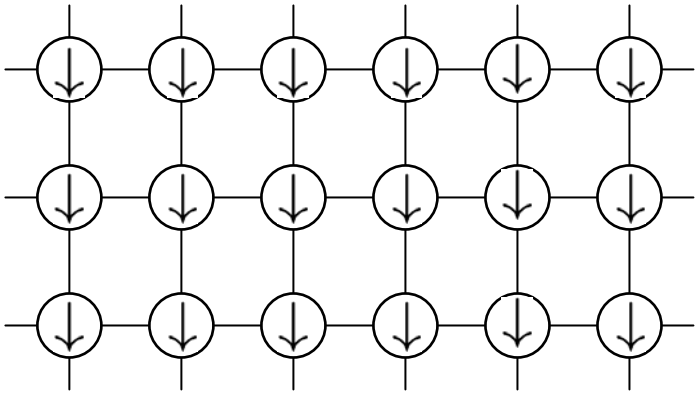
Update rule:

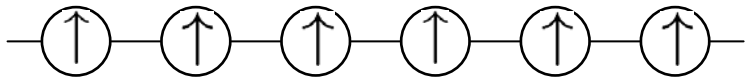
If flipping spin would decrease energy, flip spin

If flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Compare and Contrast

Encode information into the system and watch what happens



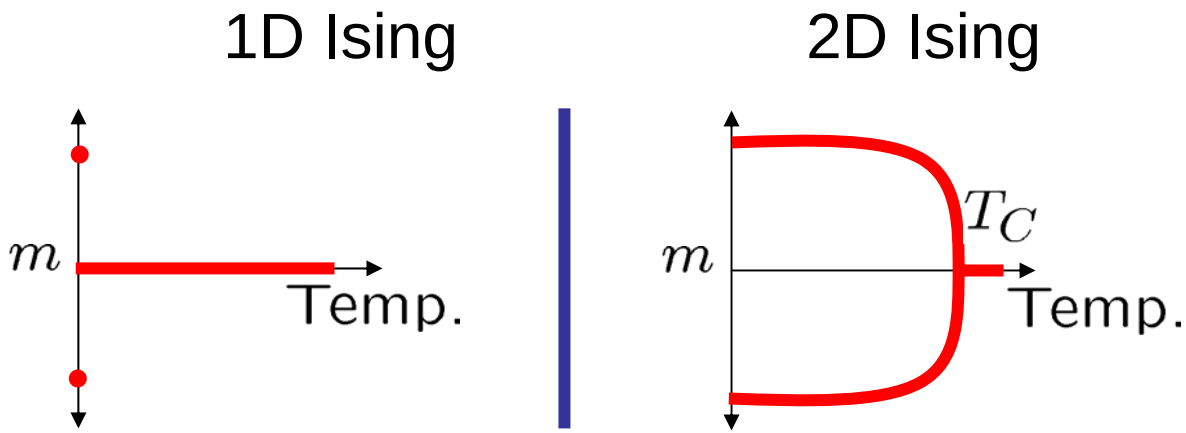


Redundancy code.

Order parameter is magnetization: $m = \frac{1}{N} \sum_i s_i$

Storage

Long time behavior (not infinite time, but effectively)



Criticism: (1) thermodynamic limit taken
(2) what if relaxation to equilibrium takes a long time?

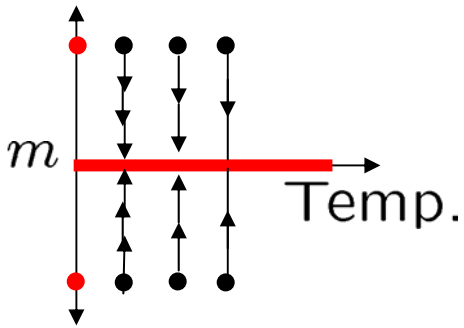
Storage



2nd attempt:

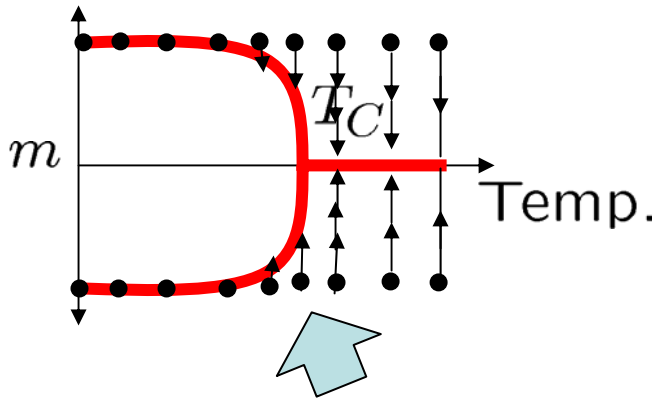
Relaxation to thermal

1D Ising



decay rate $\propto e^{-\beta J}$

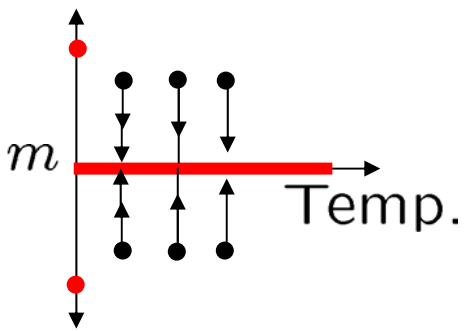
2D Ising

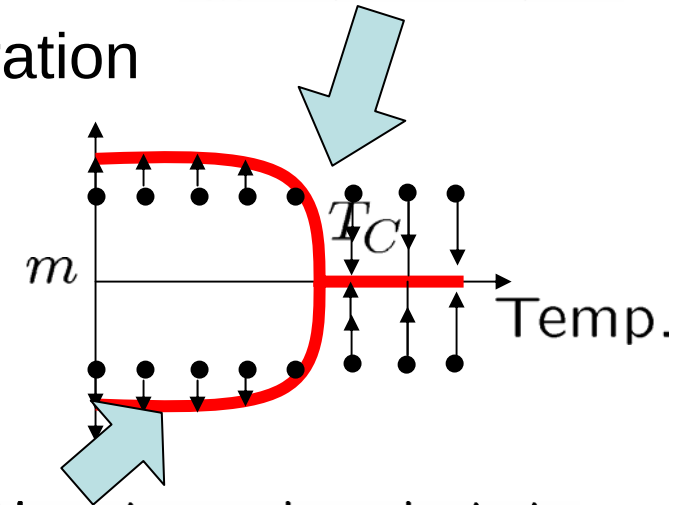


critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

Imperfect preparation





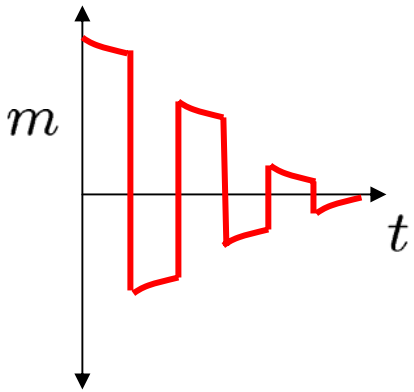
fast relaxation to ordered state

Manipulation

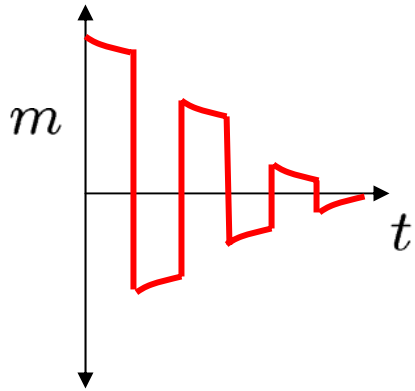


Flipping spins is imperfect

1D Ising

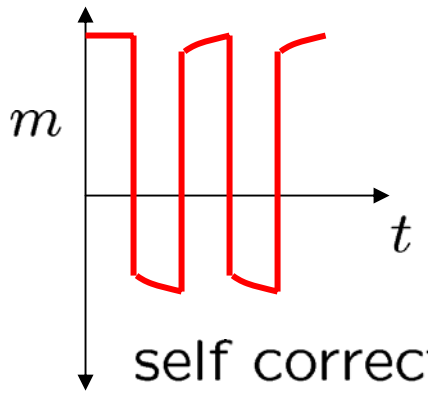


2D Ising



$T > T_C$

1D Ising
- temperature less than gap
- manipulation error must be small (i.e. NOT fault-tolerant)

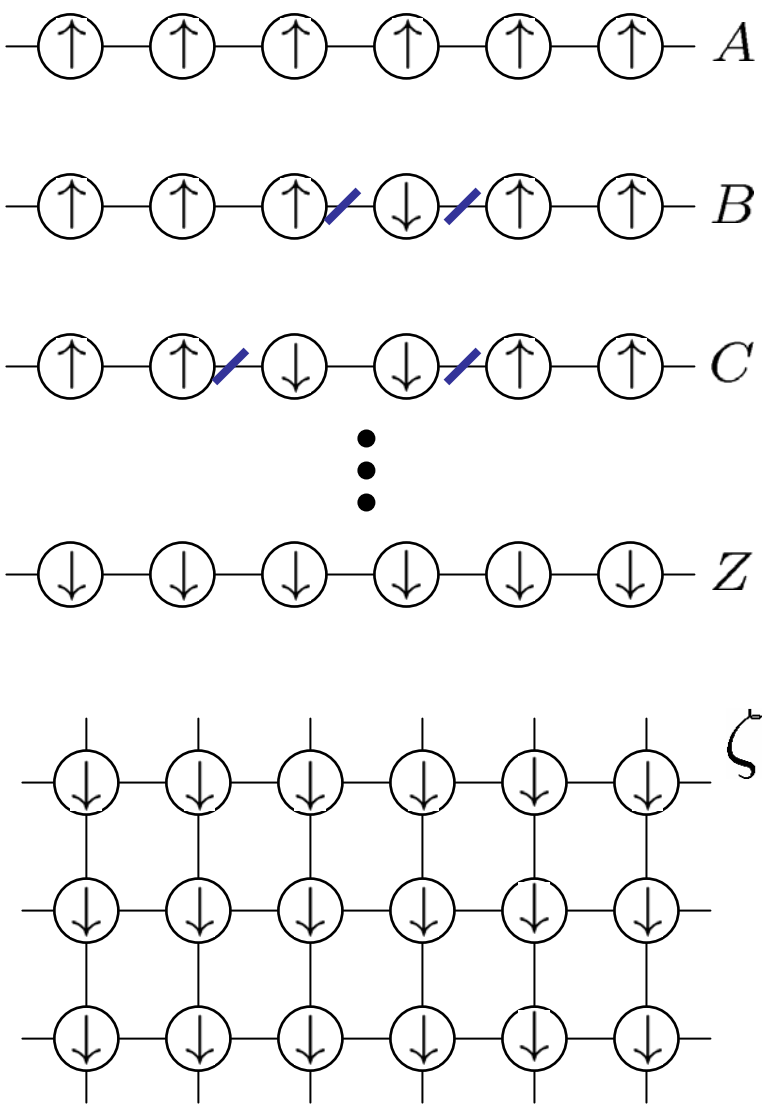


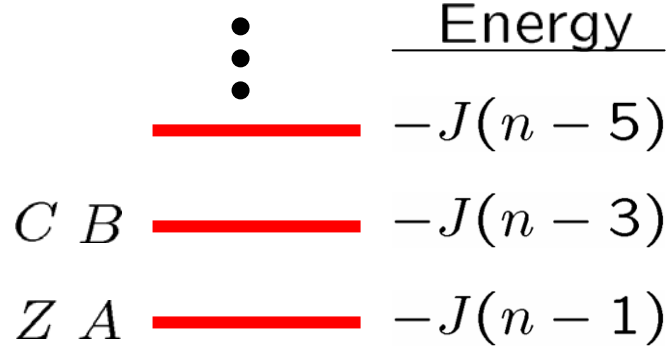
$T < T_C$

self correcting

2D Ising
- temperature less than critical temp

“Self Fixing”

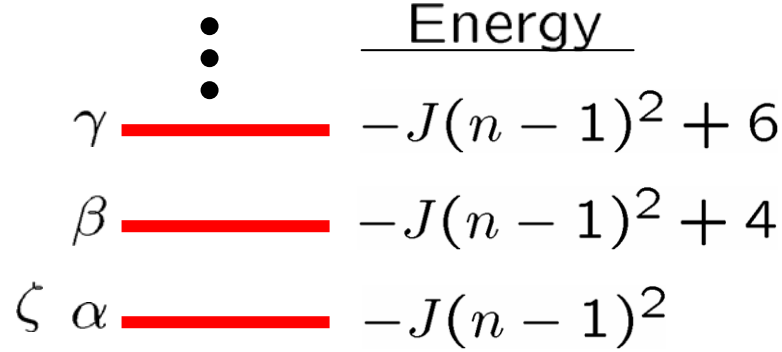




Excitations point-like

Energy to err is constant

Disordering entropy >> Energy



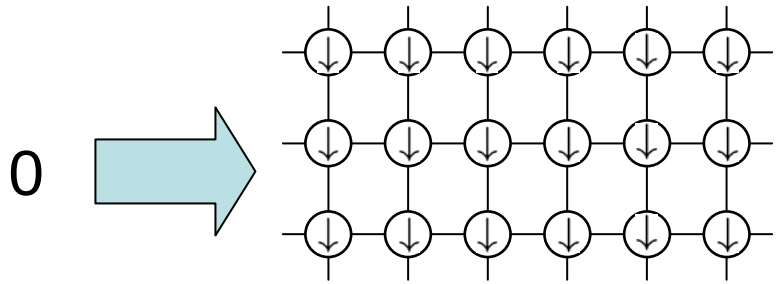
Excitations string-like

Energy to error is prop to size

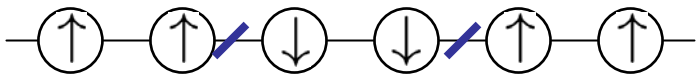
Energy >> Disordering entropy

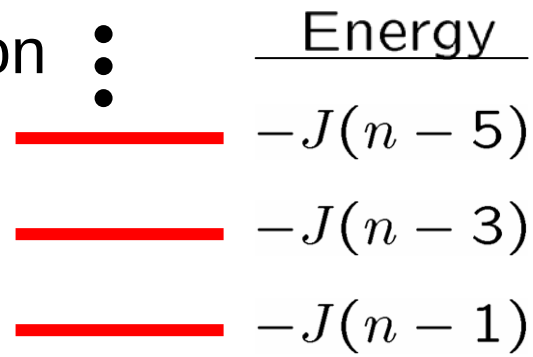
Lessons

1. encode information into error correcting code



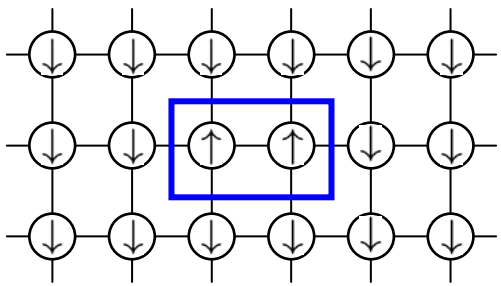
2. energy gaps can slow loss of information
   important for memory



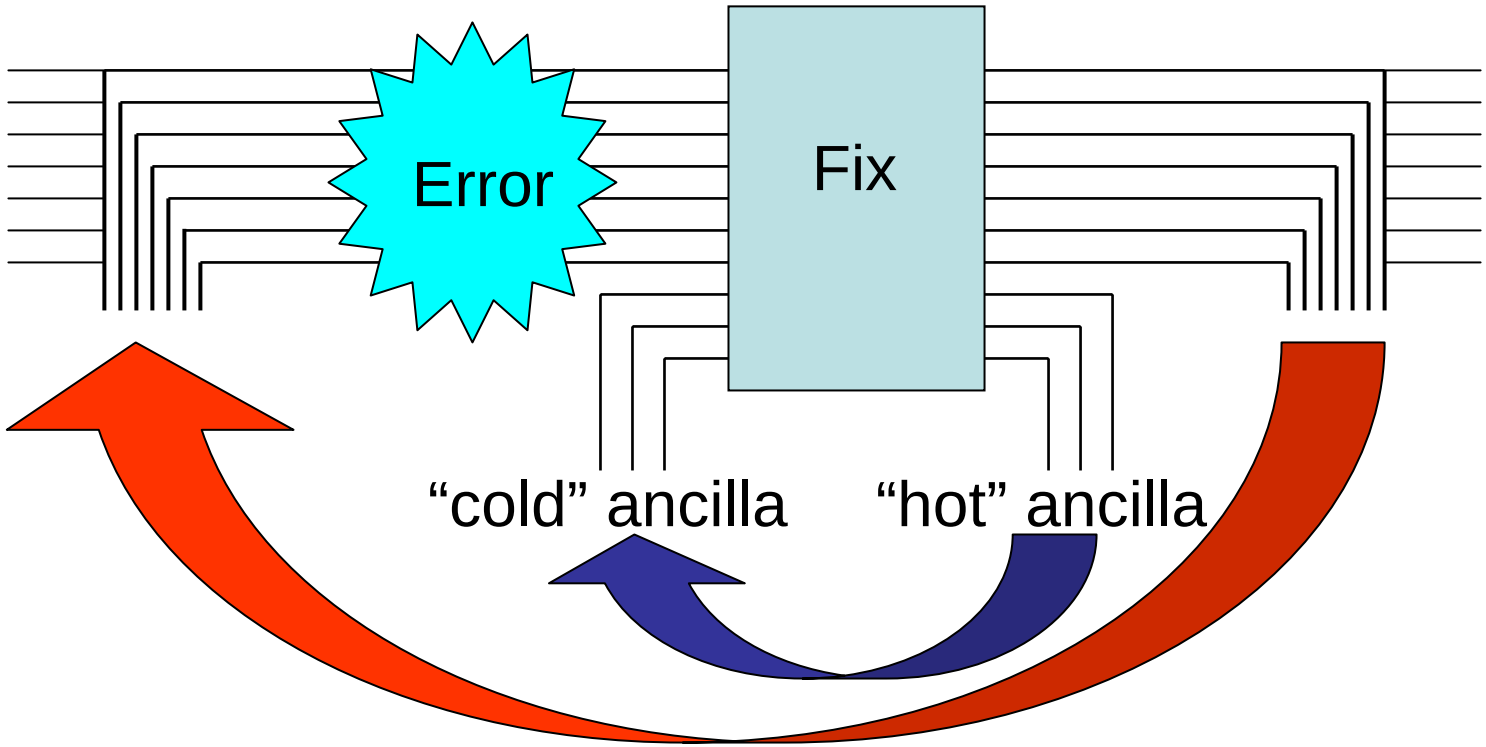


   keep cold

3. excited state energetics can “self-correct”
   important for fault-tolerant operations



Quantum Error Correction

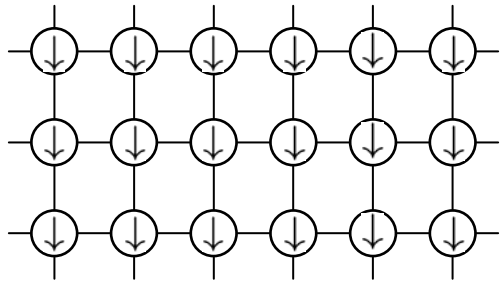


By encoding quantum information into a subspace of many quantum systems we can protect this information from noise.

Lessons Learned?

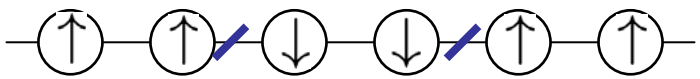
What do we do in fault-tolerant quantum computing?

✔ 1. encode information into error correcting code

0 

use information encoded into a quantum error correcting code

2. energy gaps can slow loss of information
important for memory **NO**



✔ keep cold

3. excited state energetics can “self-correct”
important for fault-tolerant operations **NO**

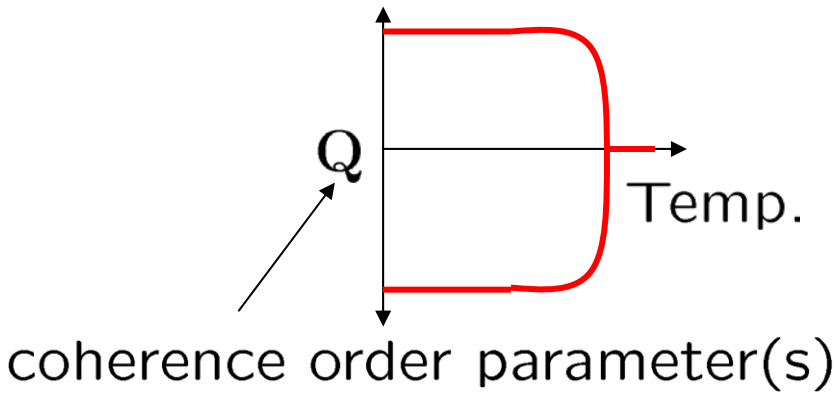
[Joked Deleted Due To Time Constraints]

The Quantum Hard Drive?

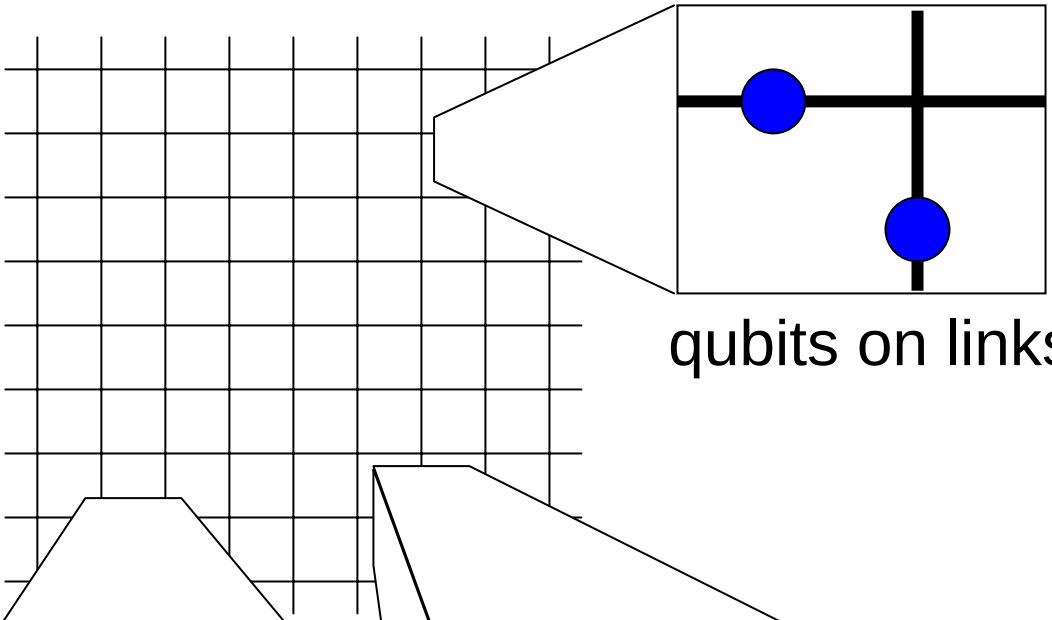


Kitaev

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



Kitaev's Model



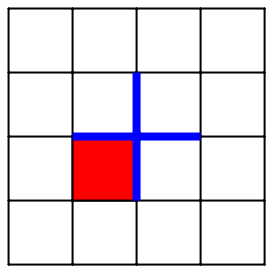
qubits on links

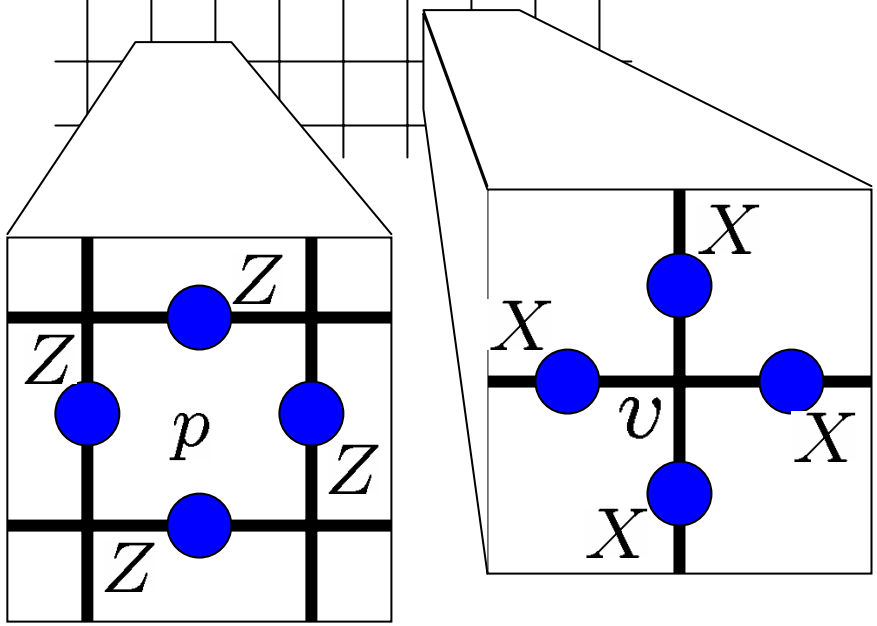
plaque operators:

$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$





vertex and plaque operators all commute with each other

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Kitaev's Model, Continued

Hamiltonian

$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

plaque operators:

$$S_p = \prod_{e \in p} Z_e$$

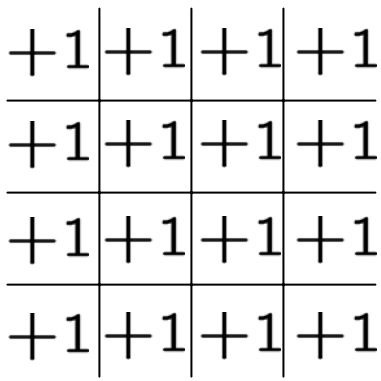
vertex operators:

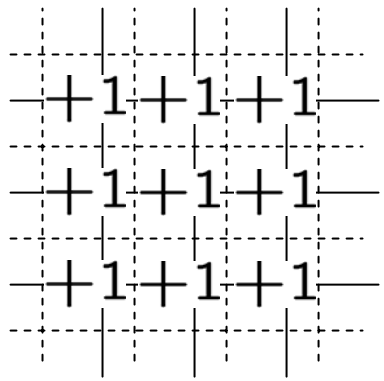
$$S_v = \prod_{e \in v} X_e$$

$$S_p|\psi\rangle = s_p|\psi\rangle \quad s_p \in \{-1, +1\}$$

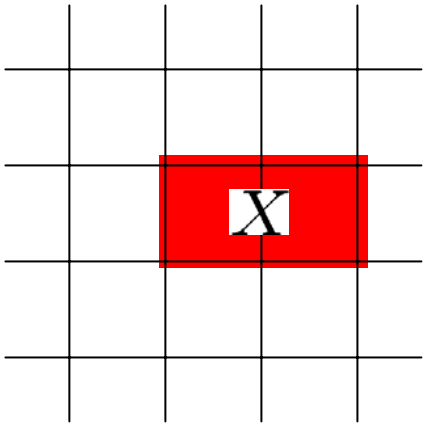
$$S_v|\psi\rangle = s_v|\psi\rangle \quad s_v \in \{-1, +1\}$$

ground state has all $s_p = s_v = +1$





Kitaev's Model



plaque operators:

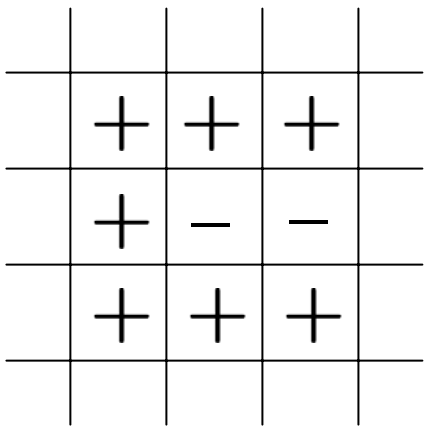
$$S_p = \prod_{e \in p} Z_e$$

vertex operators:

$$S_v = \prod_{e \in v} X_e$$

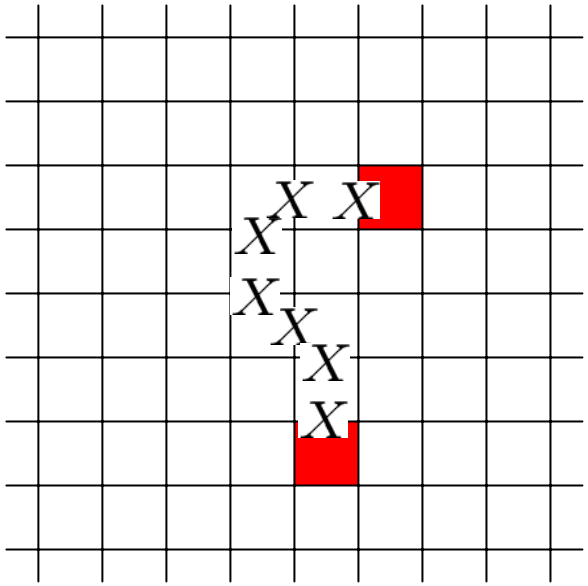
X anti-commutes with two plaque operators

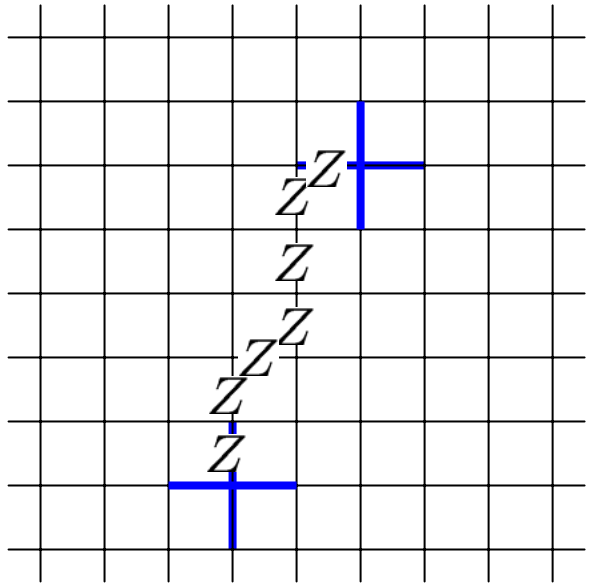
$$\{X, S_p\} = 0 \Rightarrow S_p(X|\psi_G\rangle) = -X|\psi_G\rangle$$



excitation is $4J$ above ground state

Excitations

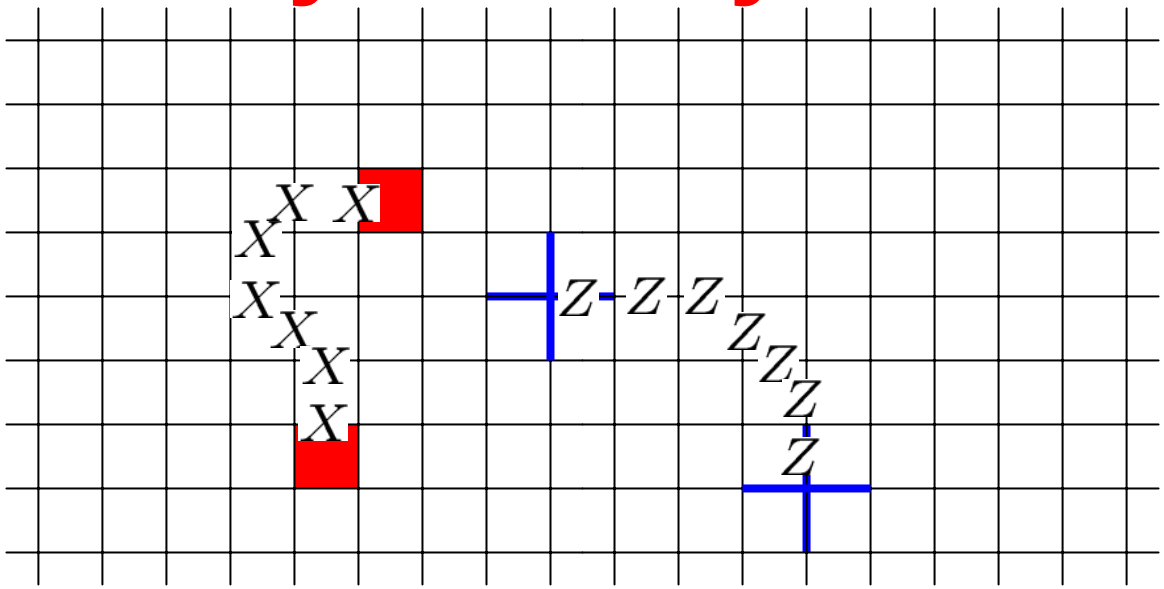




excitations particles come in pairs (particle/antiparticle)
at end of “error” chains

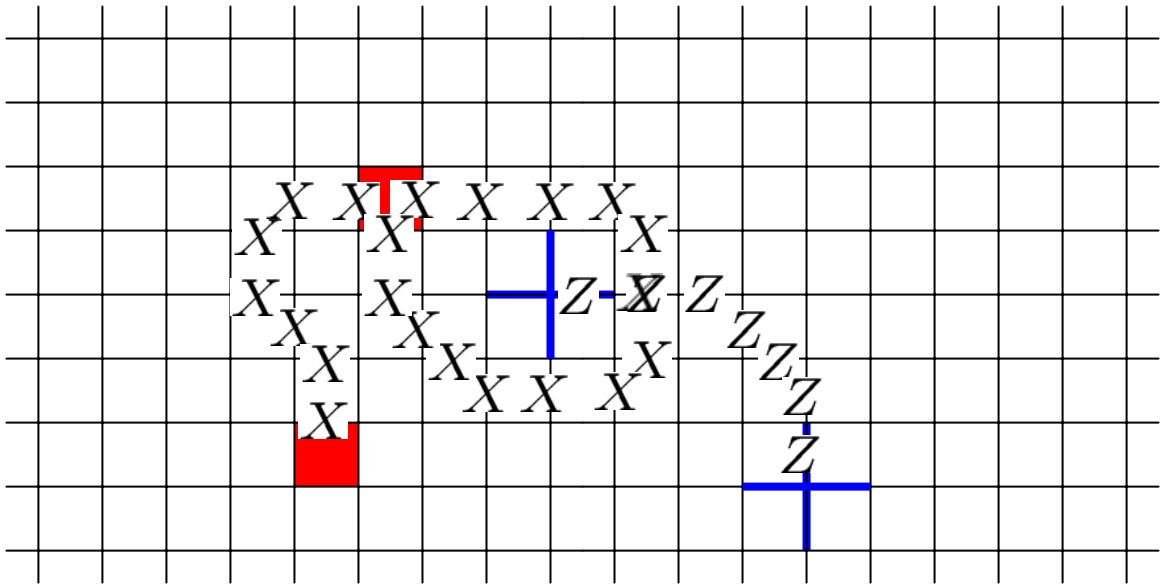
two types of particles, X-type (live on vertices of dual lattice)
Z-type (live on vertices of the lattice)

Anyons Anyone?

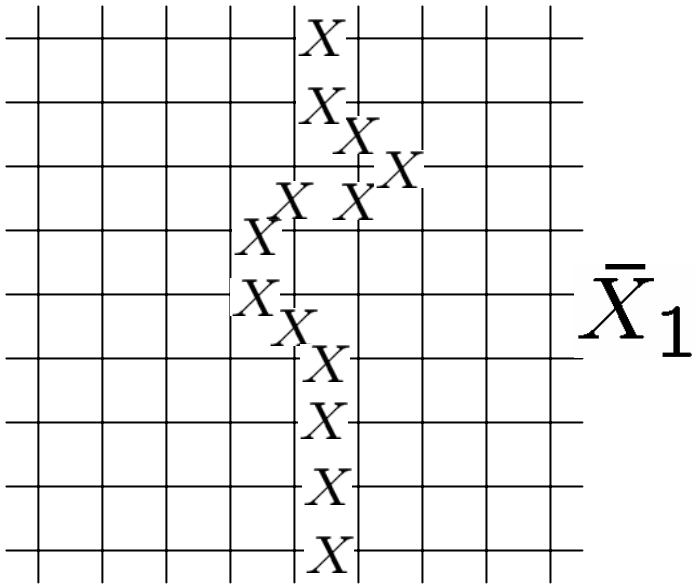


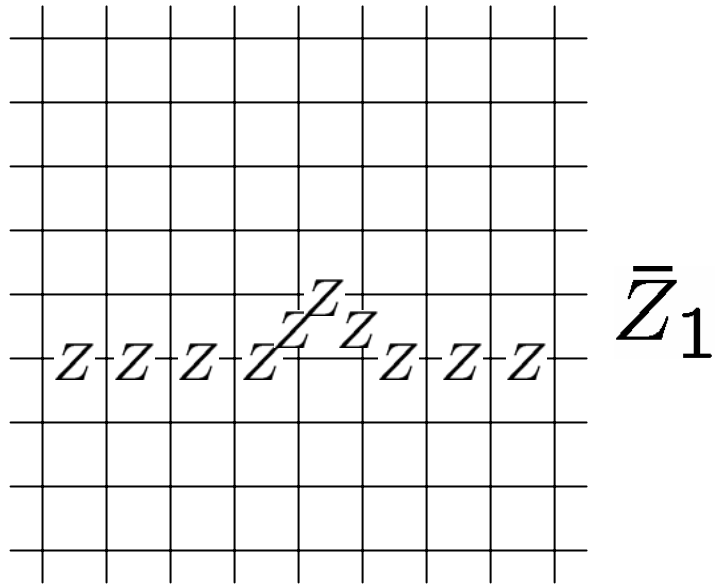
Phase:

-1



A Qubit



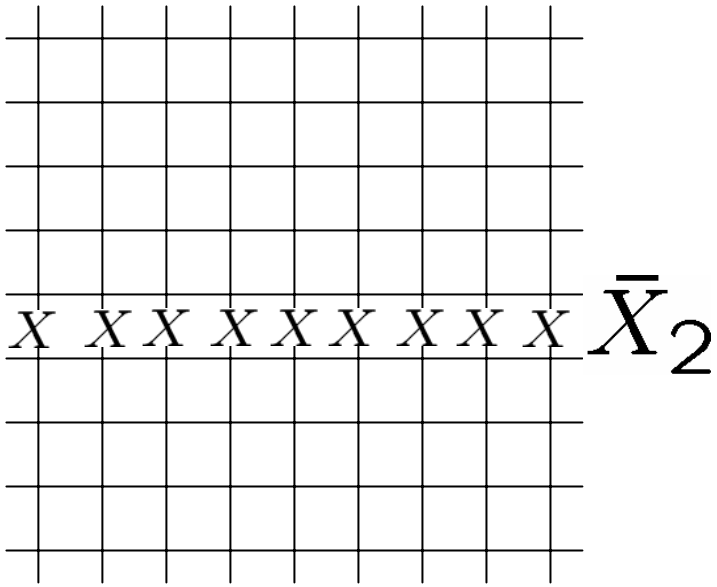


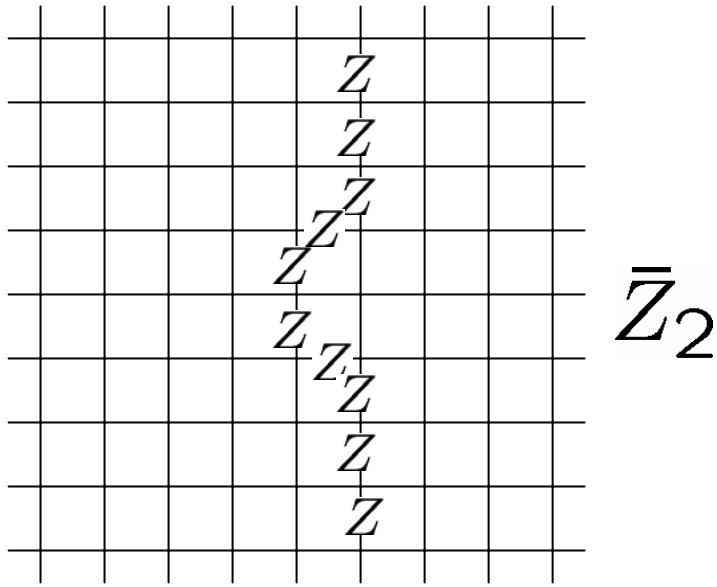
Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

$$\bar{X}_1 \bar{Z}_1 \bar{X}_1 \bar{Z}_1 = -I$$

Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Two Qubits





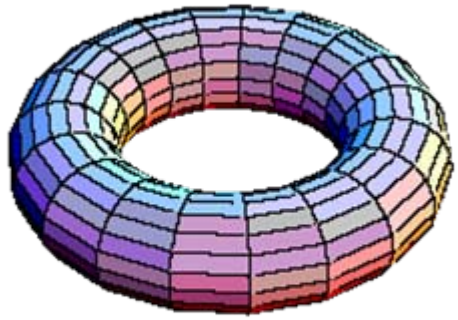
Moving an X or Z type particle on a homologically nontrivial path and annihilating with it's partner preserves the ground states

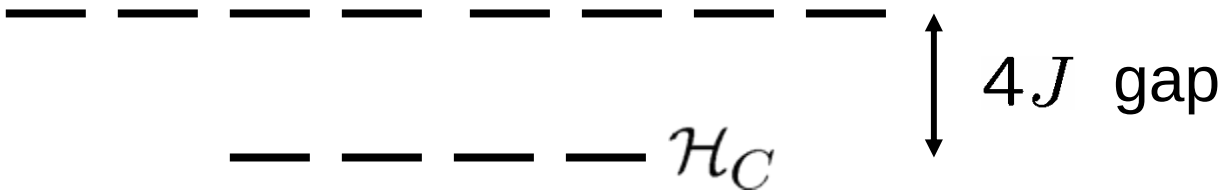
$$\bar{X}_2\bar{Z}_2\bar{X}_2\bar{Z}_2 = -I$$

Action on ground state must be representation of non-Abelian group: Acts as encoded single qubit Pauli operator.

Topological Protection

Encode two qubits
into the ground state





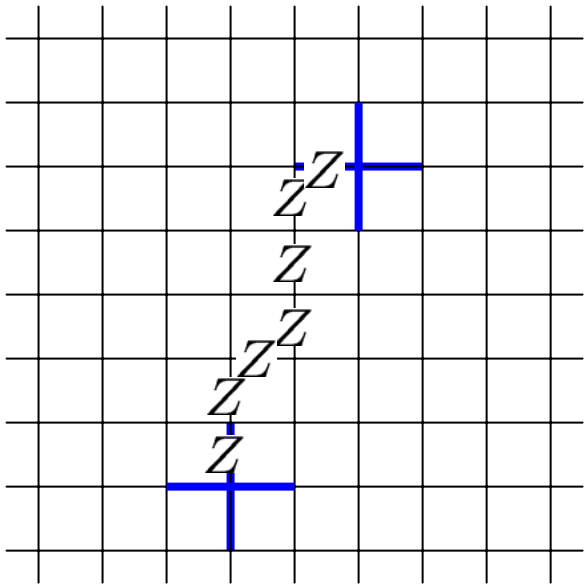
Perturbation theory:

$$V = \sum_v E_v + \sum_{vw} F_{vw}$$

But for $m \leq \left\lfloor \frac{l}{2} \right\rfloor$

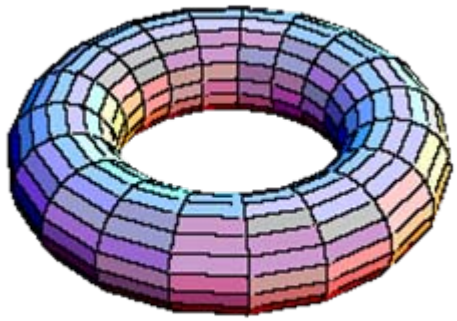
$$\langle \phi_i | V^m | \phi_j \rangle = 0$$

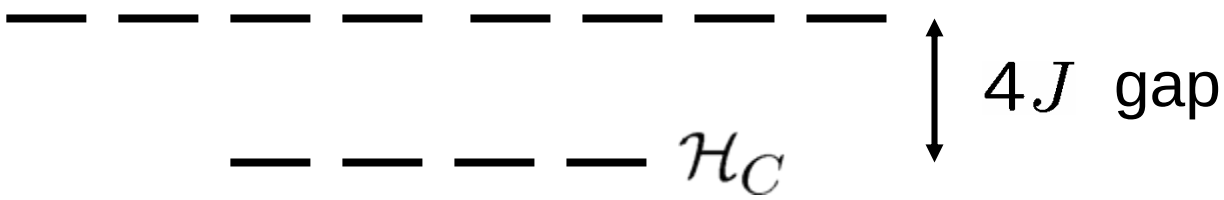
$$\langle \phi_i | V^m | \phi_i \rangle - \langle \phi_j | V^m | \phi_j \rangle = 0$$



Topological Quantum Computing

Encode two qubits
into the ground state

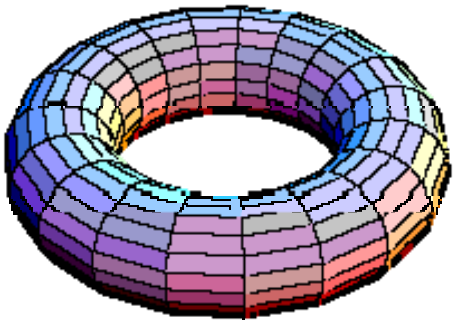




If we keep $kT \ll 4J$ then quantum information encoded into ground state will be “topologically” robust to perturbation.

We can perform gates (in this example X and Z) by creating anyon pairs and moving them around the torus.

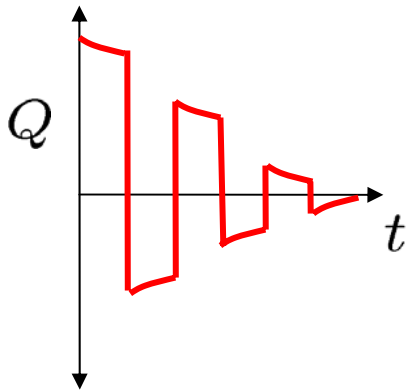
Whence Kitaev's Model?

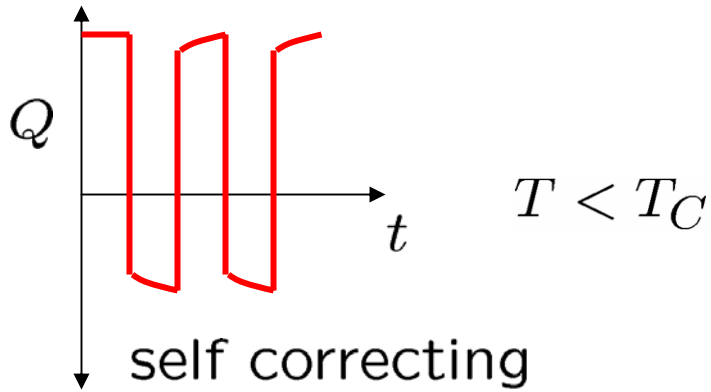


$$H = -J\left(\sum_p S_p + \sum_v S_v\right)$$

Robust storage of quantum information.

But it does not provide a mechanism for robust manipulation of quantum information.

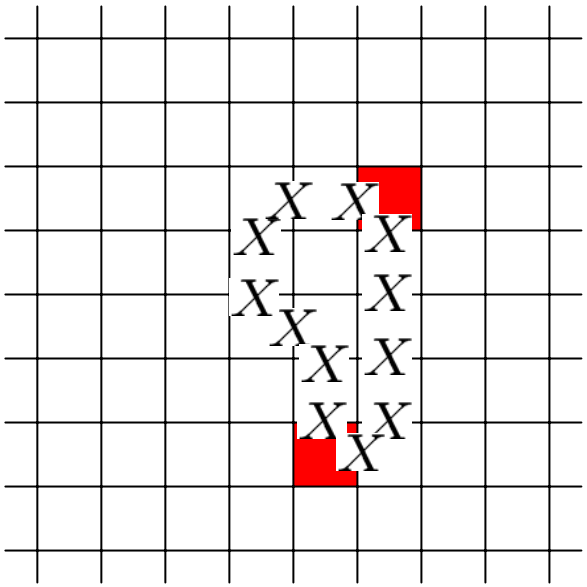




Is it possible to construct a Hamiltonian, which allows for fault-tolerant manipulation as well?

Topological Error Correction

Different mode of using Kitaev's Model, as a component in a fault-tolerant quantum computer, encode into ground state



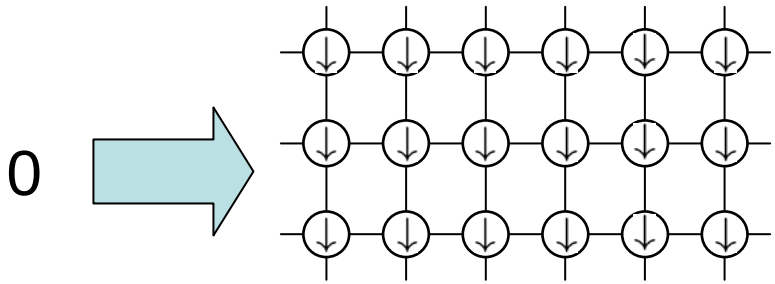
1. Errors occur
2. Measure vertex and plaque operators
3. Apply recovery operator
4. If created loop is homologically trivial, then we have restored information encoded into ground state.

Full analysis: measurements uncertain, recovery may fail, etc.
Threshold for fault-tolerant storage of quantum information:
[Dennis, Kitaev, Landahl, Preskill 2002]
[Wang, Harrington, Preskill 2003]

Lessons Learned?

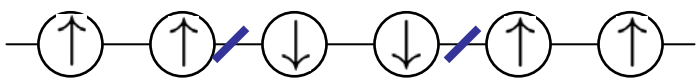
What do we do in fault-tolerant quantum computing?

✔ 1. encode information into error correcting code



use information encoded into a quantum error correcting code

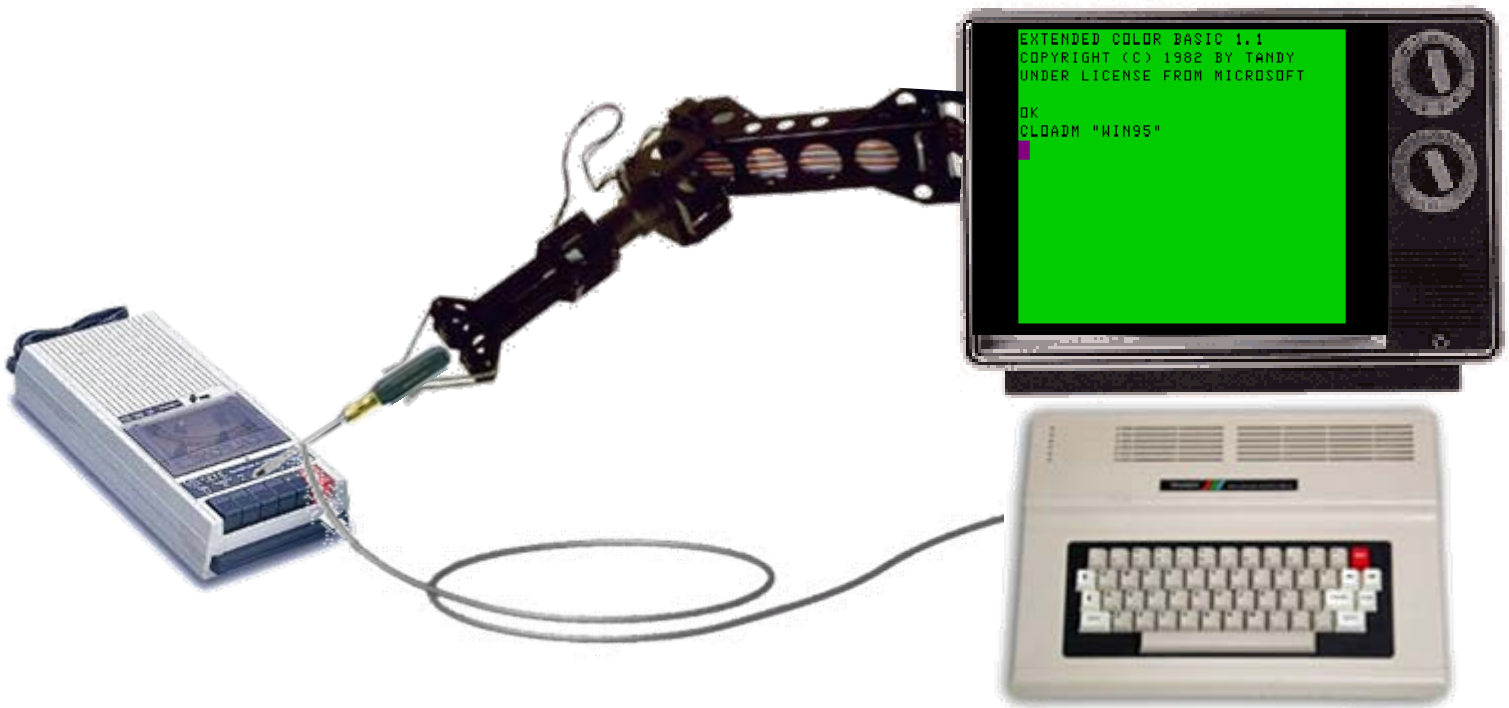
2. energy gaps can slow loss of information ✔
important for memory



✔ keep cold

3. excited state energetics can “self-correct” **NO**
important for fault-tolerant operations

Self Correcting Quantum Computers

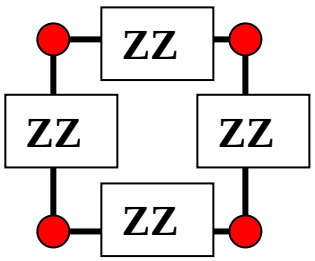


The toric code in four dimensions is probably self-correcting

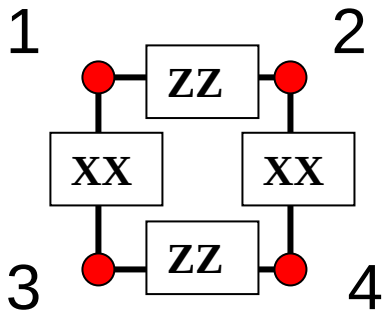
Can we self-correct in realistic systems?

D. Bacon “Operator Quantum Error Correcting Subsystems for Self-Correcting Quantum Memories”
Phys. Rev A, 73, 012340 (2006)

A Small Quantum Try



classical



quantum

Hamiltonian

$$H = -(Z_1Z_2 + Z_3Z_4 + X_1X_3 + X_2X_4)$$

quantum error detecting code

$$S_1 = Z_1Z_2Z_3Z_4$$
$$S_2 = X_1X_2X_3X_4$$

symmetry

$$[H, S_1] = [H, S_2] = 0$$

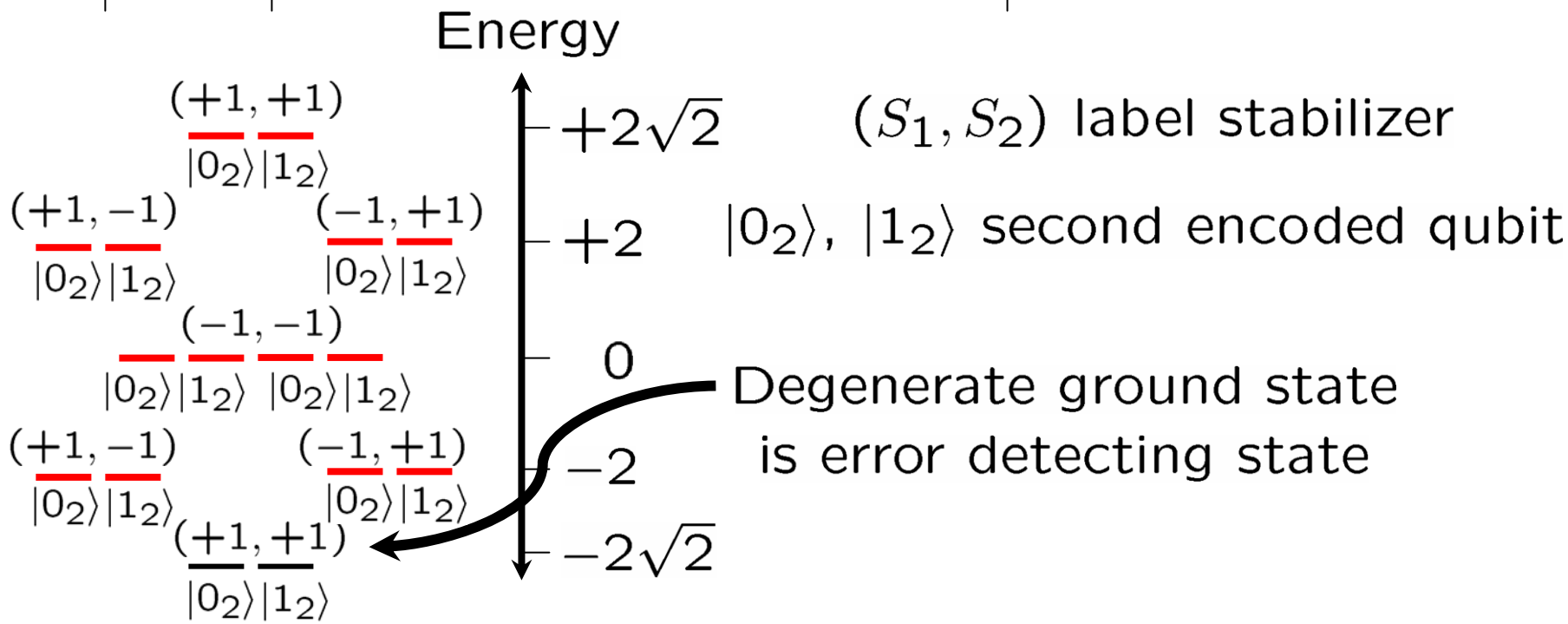
$$\bar{Z}_1 = Z_1Z_2 \qquad \bar{X}_1 = Z_1Z_3$$
$$\bar{Z}_2 = Z_1Z_3 \qquad \bar{X}_2 = X_1X_2$$

$$H = -\bar{Z}_1(I + S_1) - \bar{X}_1(I + S_2)$$

simultaneously diagonalize with symmetries

H does not depend on second qubit

S_1	S_2	$H = \bar{X}_1(1+S_1) + \bar{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\bar{X}_1 + \bar{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\bar{X}_1$	± 2
-1	$+1$	$2\bar{Z}_1$	± 2
-1	-1	0	0





Reminder



two-level system perturbatively coupled to a thermal boson bath

$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

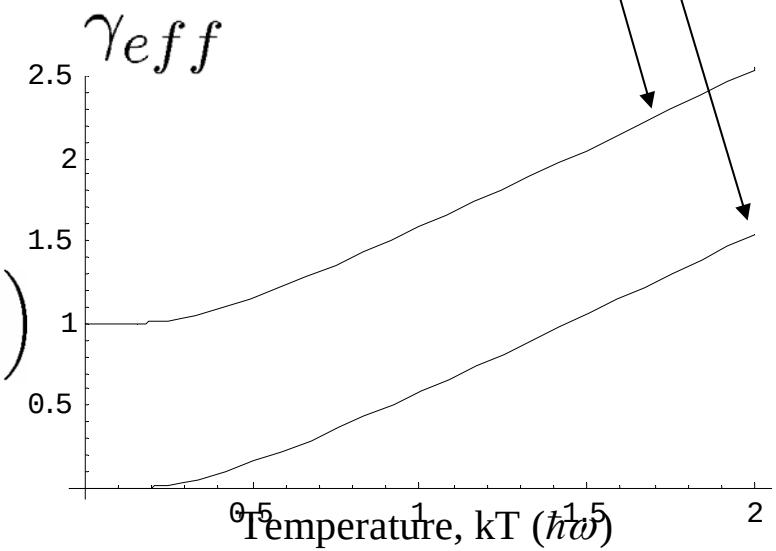
"non-dissipative"

$\hbar\omega$

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



Supercoherence

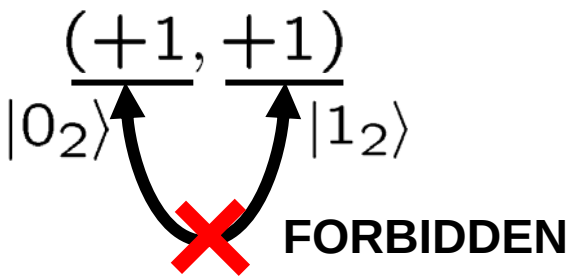
Bacon, Brown, and Whaley, Phys. Rev. Lett. **87**, 247902 (2001)

$$H = -(Z_1 Z_2 + Z_3 Z_4 + X_1 X_3 + X_2 X_4)$$

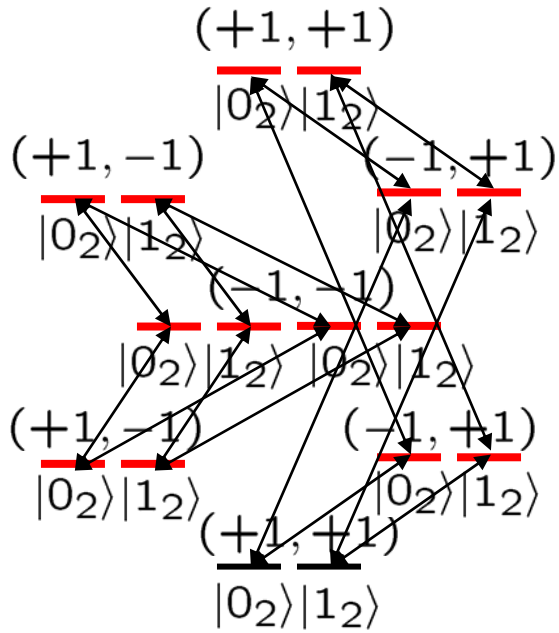
$S_i P = -P S_i$ for some i $\quad P$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S_1,S_2) and hence take ground state to higher energy level.



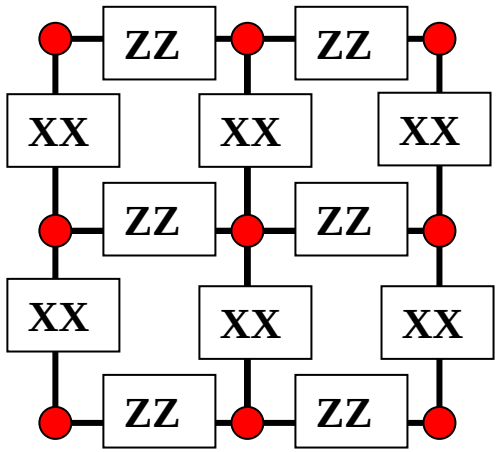
Effect of **ZIII**



To first order all single qubit decoherence becomes heating

Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$

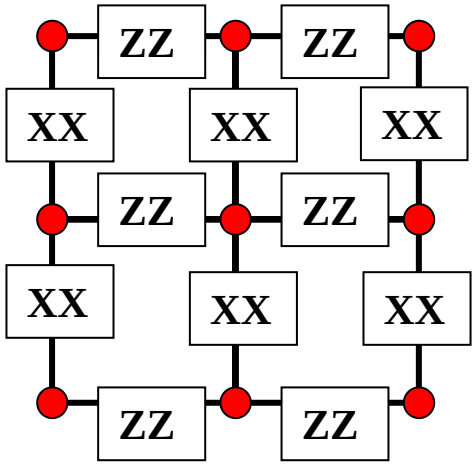


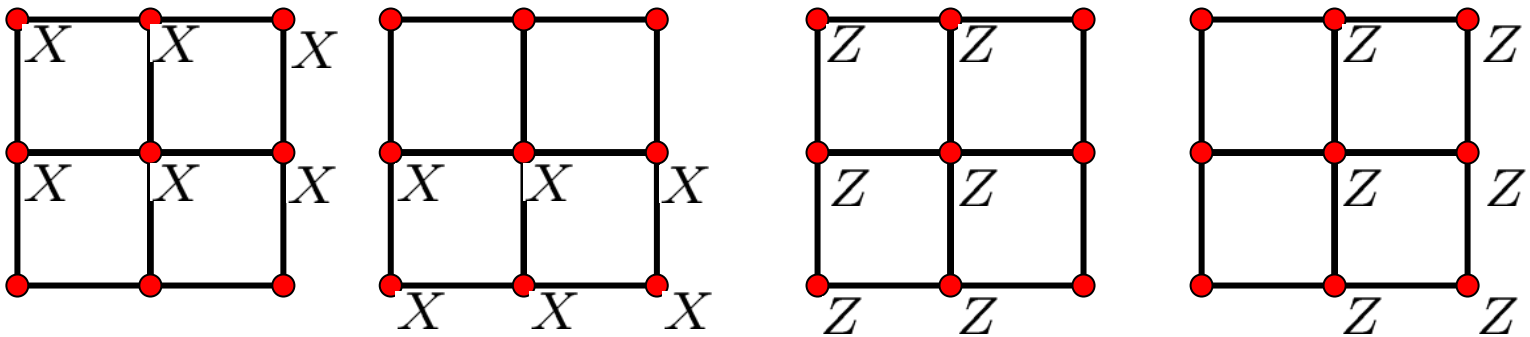
Ground state is a new kind of quantum error correcting code!

quantum error correcting subsystem

Quantum Compass Model

$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$





Abelian symmetries

Subspaces Versus Subsystems

The most general way to encode quantum information is into subsystem (not subspaces)



two qubits

$\mathcal{H}_2 \otimes \mathcal{H}_2$

encode a qubit in a subspace

$$|0_L\rangle = |1\rangle \otimes |1\rangle$$

$$|1_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$$

$$\alpha|0_L\rangle + \beta|1_L\rangle$$

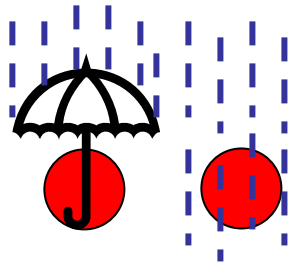
encode a qubit in a subsystem

$$(\alpha|0_L\rangle + \beta|1_L\rangle) \otimes |\psi\rangle$$

generic subsystem:

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

Subsystem Error Correction



Only need to protect subsystem, not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

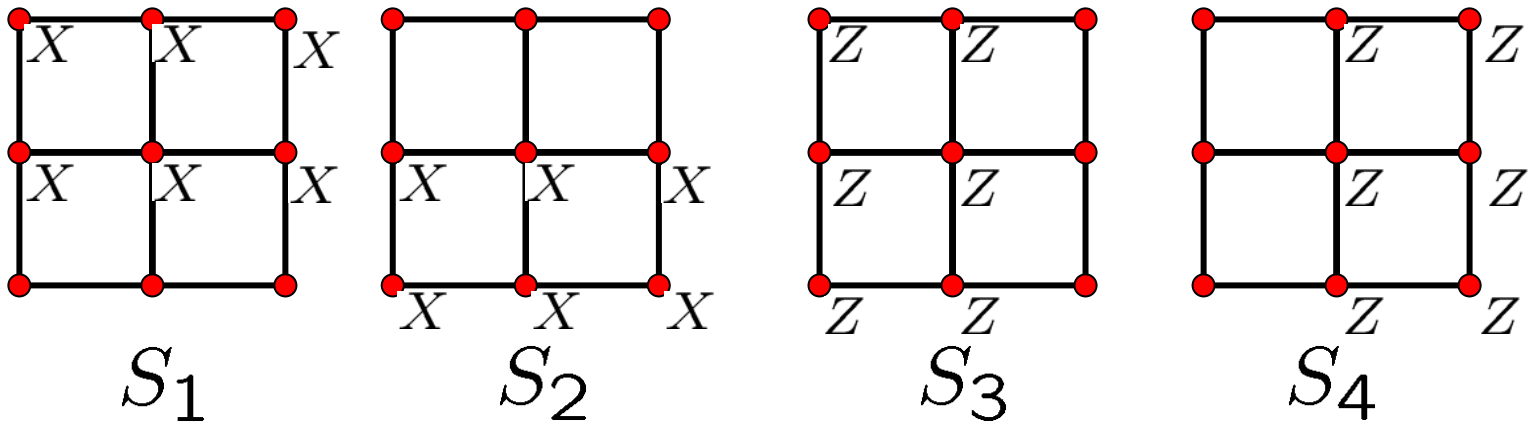
$$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0 \quad \mathcal{E} \quad \mathcal{R} \quad (|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$$

encoded quantum information

error recovery

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Quantum Subsystem Code

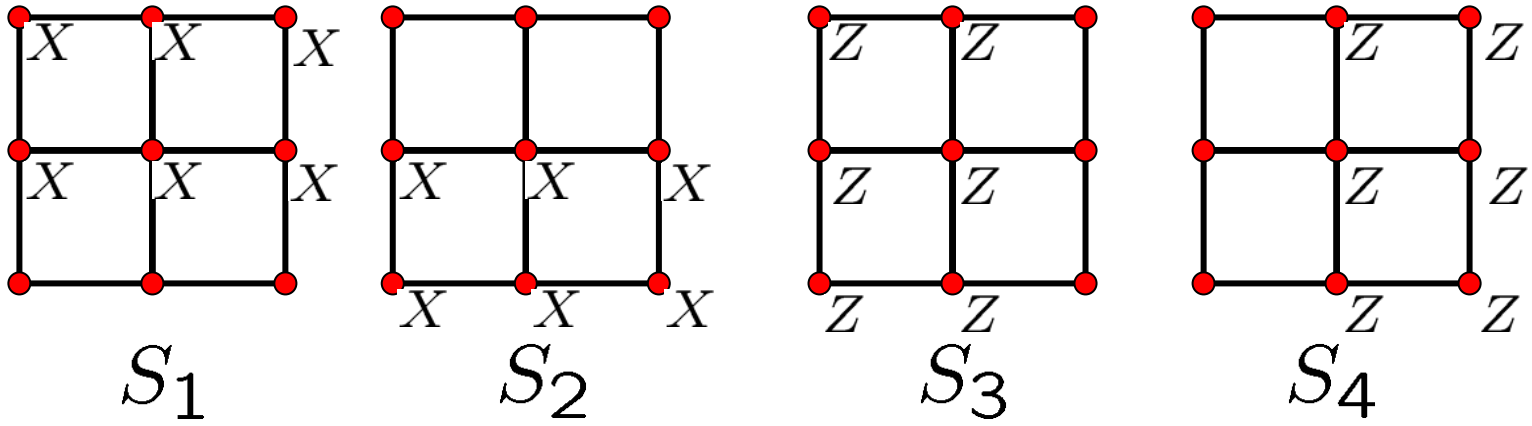


1. Measure these operators
2. Diagnose parity of bit and phase flips in column and rows
3. Apply recovery routine for parity of errors

Routine fixes error UPTO an operation generated by bonds of Hamiltonian

$$X_{i,j}X_{i+1,j}, Z_{i,j}Z_{i,j+1}$$

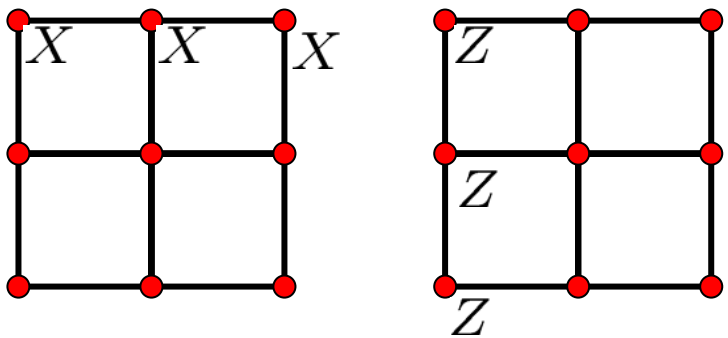
Quantum Subsystem Code



S_1 S_2 S_3 S_4

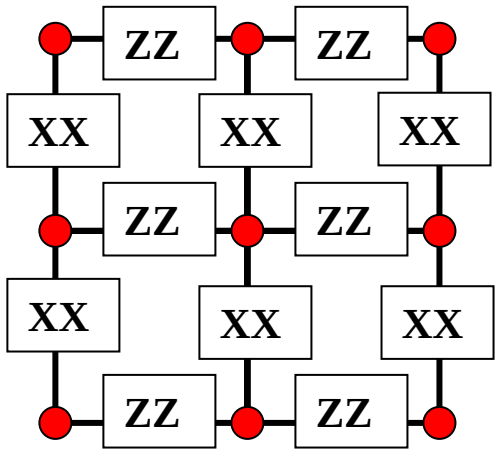
$$X_{i,j}X_{i+1,j}, Z_{i,j}Z_{i,j+1}$$

extra symmetry which implies states are doubly degenerate:



Quantum Compass Model

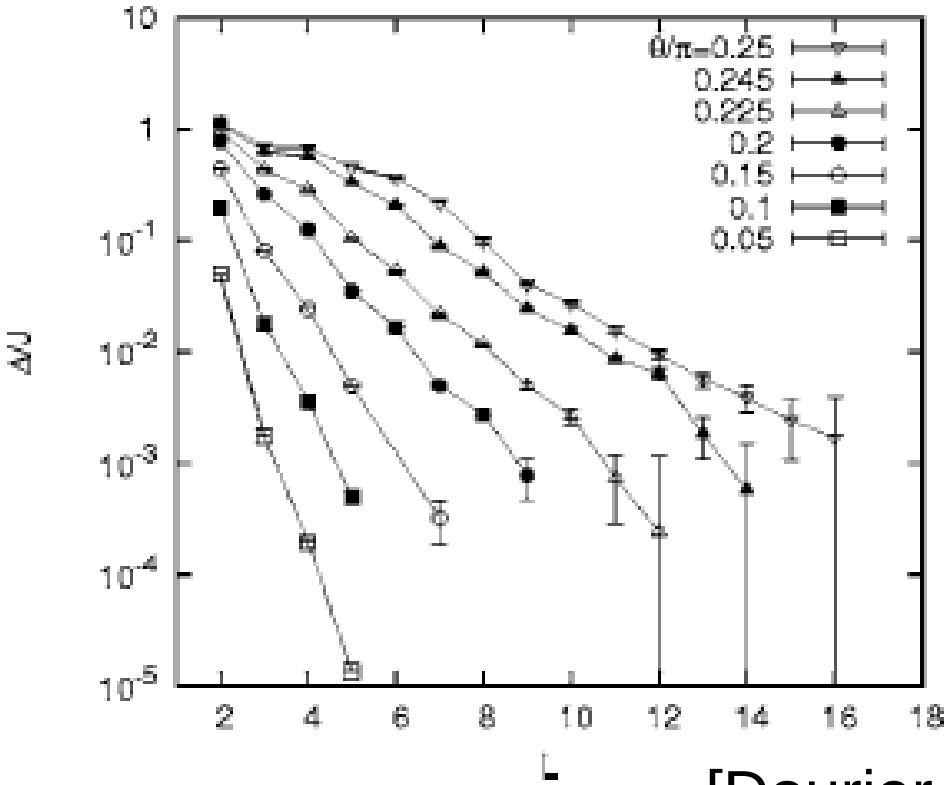
$$H = -J_X \sum_{ij} X_{i,j} X_{i+1,j} - J_Z \sum_{ij} Z_{i,j} Z_{i,j+1}$$



Ground state is a new kind of quantum error correcting code!

quantum error correcting subsystem

Problems:

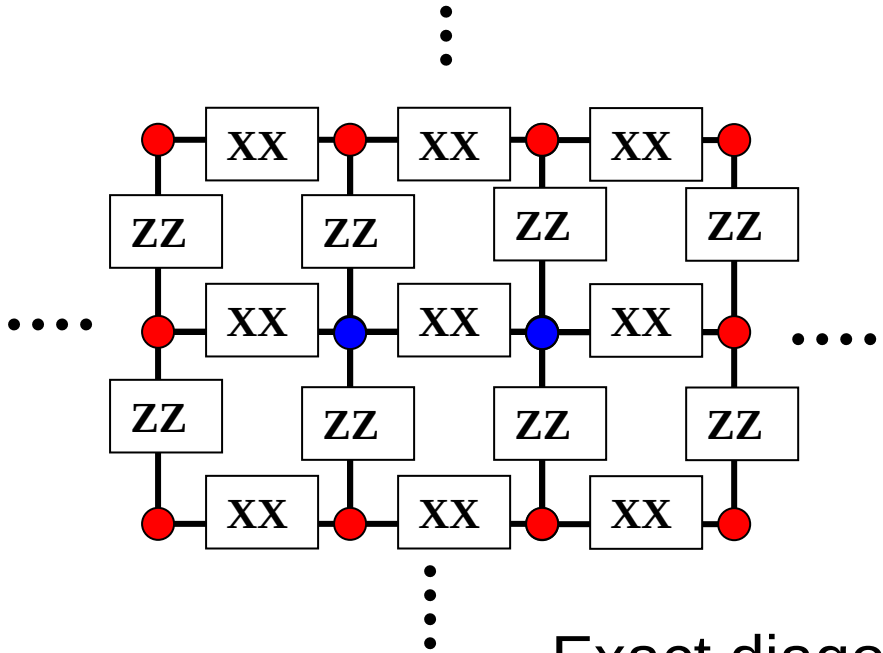


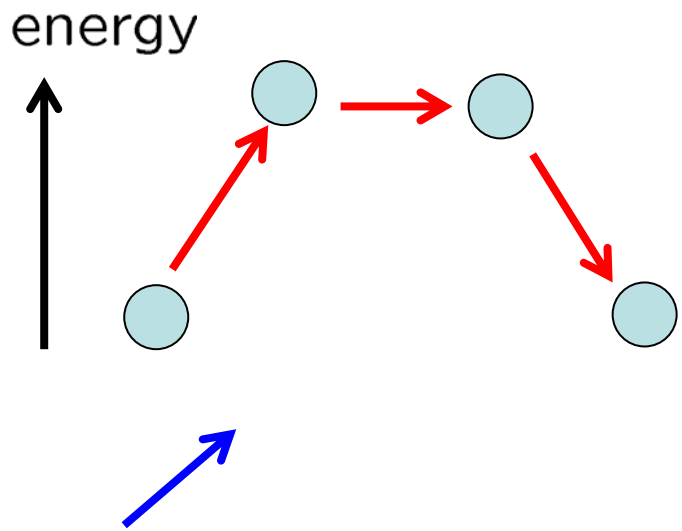
[Dourier, Becca, Mila 05]

$$H = -J \sum_{i,j=1}^{n} \cos(\theta) X_{i,j} X_{i+1,j} + \sin(\theta) Z_{i,j} Z_{i,j+1}$$

no gap in the thermodynamic limit

Problems, Continued



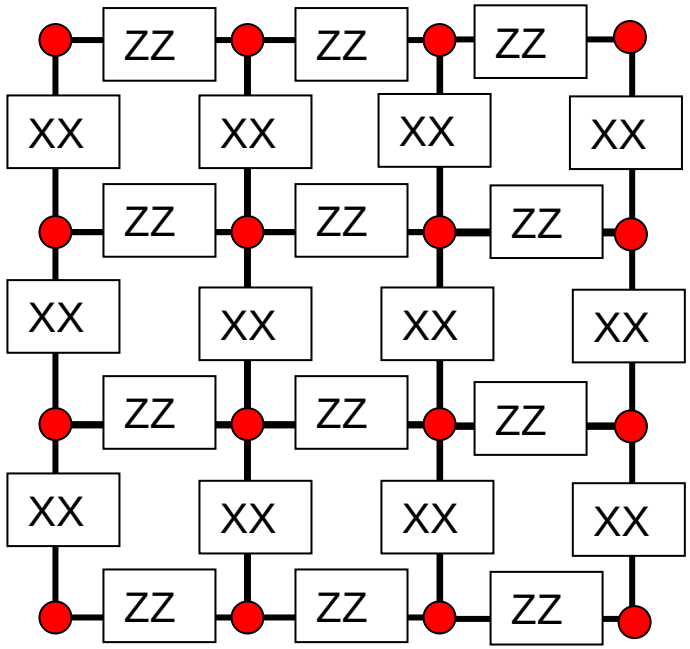


Exact diagonalization 3x3, 4x4, 4x5

Even at finite size, not “self-correcting”

Toward Self-Correction

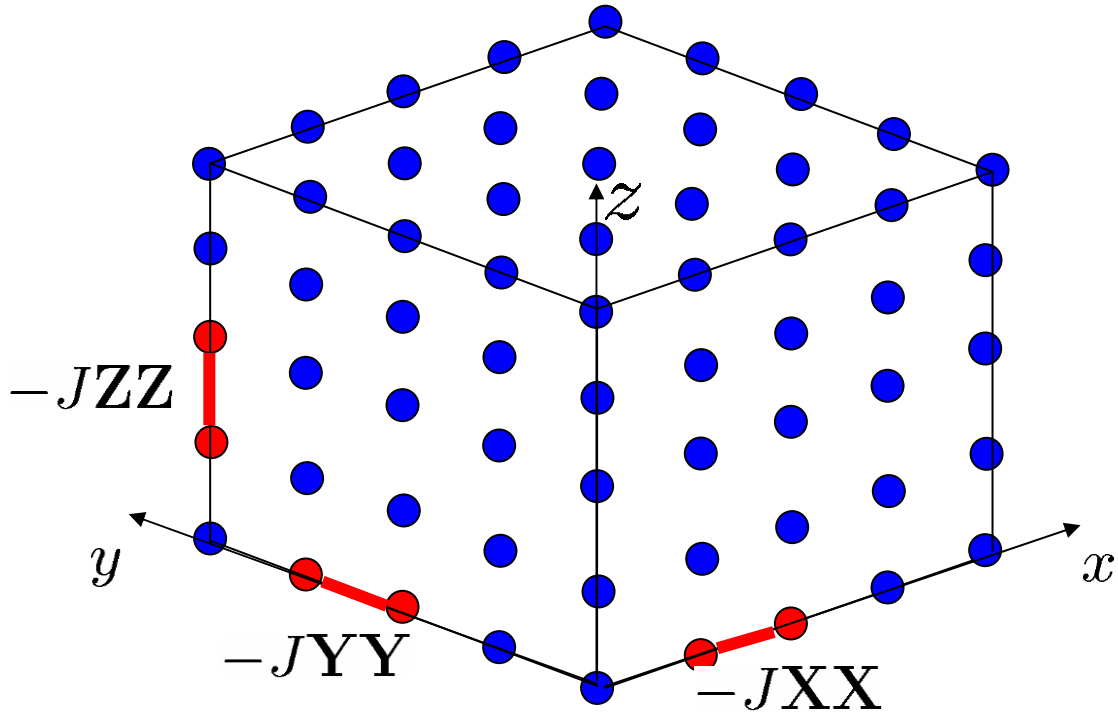
n by *n* square lattice



$$H = -\sum_{i,j=1}^{n}\sum_{k=1}^{n-1} J_X(k) X_{i,j} X_{i+k,j} + J_Z(k) Z_{i,j} Z_{i,j+k}$$

Exact diagonalization of 4x4 model, shows that with constant couplings, this model is self-correcting!

Toward Self-Correction



$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

The Future

Can techniques used in creating two qubit interactions be used to engineer such interactions
(Ideas from quantum simulation? Deng, Porras, and Cirac?)

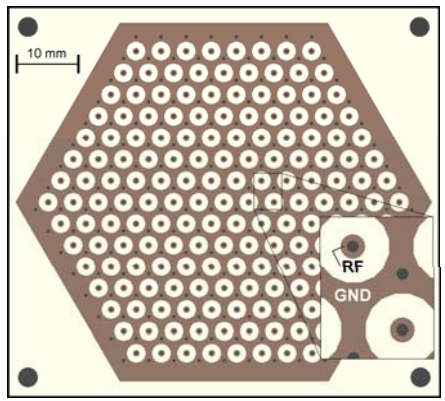


Figure 2. Hexagonal planar ion trap. 2D mult-zone ion traps allow one to trap ions in any spatial arrangement including a disordered lattice.

Picture courtesy Ken Brown
(Isaac Chuang's Lab, MIT)

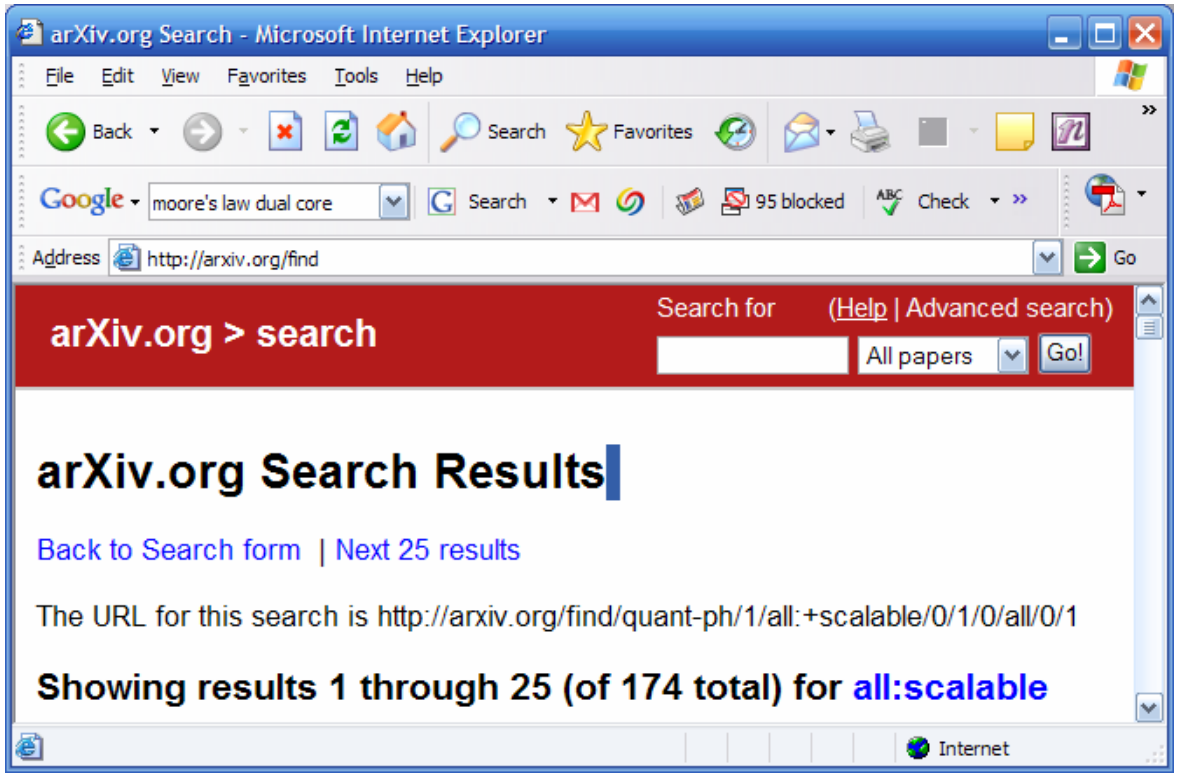
A Different Path



Quantum error correction can be performed by cooling an interacting quantum system.

Potential for robust substrates for quantum information.

What Does “Scalable” Mean?



1. All of the necessary components.
2. Easy to fabricate.
3. Below “threshold”

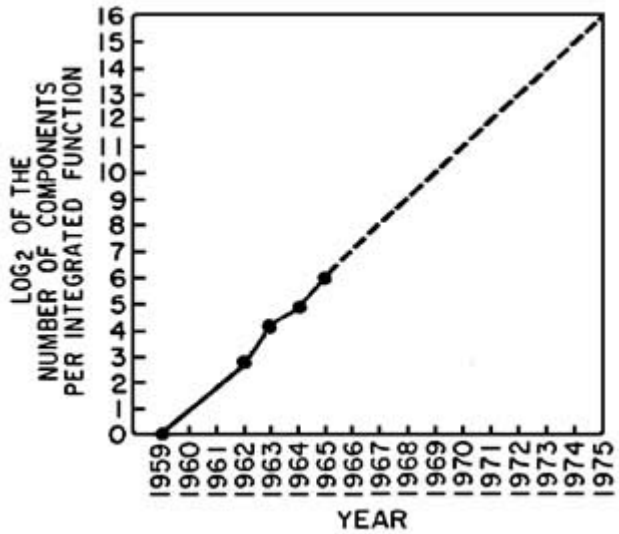
1+3. All of the necessary components which operate below “threshold”

1+2+3. All of the necessary components, below “threshold”, and easy to fabricate.

What Does “Scalable” Mean?



Gordon Moore



Scalable has a rate and a lifetime.

Scalable means that you can predict from current technology this rate and a lower bound on the lifetime.

Scalable means that you can write Moore’s 1965 paper

….and be right!

The Future

Is it possible to engineer the systems described above?
(Ideas of Deng, Porras, and Cirac?)

Possible to engineer such systems in planar ion trap system?

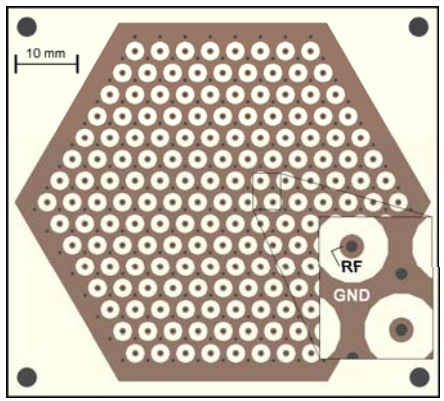


Figure 2. Hexagonal planar ion trap. 2D mult-zone ion traps allow one to trap ions in any spatial arrangement including a disordered lattice.

Picture courtesy Ken Brown
(Isaac Chuang's Lab, MIT)

Quantum Architecture?



Chuang (MIT), Chong (UCSB), Kubiatowicz (UCB), Oskin (UW)

Some early work: Oskin, Chong, Chuang. IEEE Computer, Jan. 2002
“A Practical Architecture for Reliable Quantum Computers”

Quantum Architecture?

